

of right singular vectors of $\begin{bmatrix} S_x & S_y \end{bmatrix}$. Here, V_{11} and V_{22} are $n \times n$. The ESPRIT estimates of the values $e^{i\omega_0 \|\Delta\|_2 \sin(\theta_k)/c}$ are the eigenvalues λ_j of $-V_{12} V_{22}^{-1}$ so that the estimate $\hat{\theta}_j$ of θ_j is

$$\hat{\theta}_j = \sin^{-1}(c \arg(\lambda_j) / (\|\Delta\|_2 \omega_0))$$

for $j = 1, 2, \dots, n$.

Examples:

1. We consider a single signal arriving at an array of two sensors with array response vector

$$\mathbf{a}(\theta) = \begin{bmatrix} e^{-i(\pi/2-\theta)} \\ 1 \end{bmatrix}.$$

If observations of the sensor outputs and solution of the generalized eigenvalue problem $R_x - \lambda R_n$ suggest that

$$X_s X_s^* = \begin{bmatrix} i/\sqrt{3} \\ 2/\sqrt{3} \end{bmatrix} \begin{bmatrix} -i/\sqrt{3} & 2/\sqrt{3} \end{bmatrix}$$

is the projection for the signal subspace, then the projection for the noise subspace is

$$X_n X_n^T = \begin{bmatrix} -2i/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 2i/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}.$$

In applying the MUSIC algorithm, we have

$$M(\theta) = \frac{2}{3|2e^{i\theta} + 1|^2}.$$

This function has a maximum of $2/3$ at $\theta = \pi$. Thus, the MUSIC estimate of the angle from which the signal arrives is $\theta = \pi$.

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Applications to Geometry

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Geometry

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Many topics taught in an introductory linear algebra course are often motivated by reference to elementary geometry. The geometry is treated as something the student is already familiar with, and reference to it is made to give the student a better understanding of algebraic concepts that may be considered, at least on first acquaintance, to be rather abstract. What is interesting and important is that the process can be reversed: Assuming the linear algebra as known, geometric concepts can be defined and developed on a rigorous basis. The use of vector methods in geometry often provides new ways of looking at old problems and also helps demonstrate deep and beautiful connections between algebra and geometry.

This chapter begins by discussing affine geometry, which can be defined, very roughly, as Euclidean geometry without any mention of measurement. Thus, affine theorems concern such things as incidence and parallelism. An affine space is defined in terms of an action of a vector space V on a set X , pursuant to which a vector $\mathbf{v}$ acts on a point A of X by sending it to another element $\mathbf{v}(A)$ of X , and two points A and B of X define a unique vector $\overrightarrow{AB}$, which acts on A by sending it to B . It might be helpful for the reader to think of $\overrightarrow{AB}$ as an arrow starting at point A and ending at point B . This vector then acts on an arbitrary point C of X by placing the starting point of the arrow at C and letting the end point of the arrow be $\mathbf{v}(C)$.

By introducing an inner product on a real affine space, Euclidean geometry can be defined. Actually, a very interesting mathematical theory can be developed by considering, more generally, an arbitrary bilinear form, but in the interest of simplicity, only (positive definite) real inner products are considered in this chapter. The general theory is discussed in Chapter III of [Art57] and Chapter 2 of [ST91].

Finally, projective geometry is considered. Here the points of the geometry are not vectors but one-dimensional subspaces and the resulting incidence properties are different than for Euclidean geometry. Projective geometry plays an important role in (among other things) the mathematical theories of elliptic curves and algebraic geometry, but these topics are far beyond the scope of this chapter.

Throughout this chapter, F is a field and V is a finite dimensional vector space over F .

65.1 Affine Spaces

Definitions:

An **affine n -space** is an ordered pair (X, V) where X is a nonempty set and V is an n -dimensional vector space over a field F , which acts on X in the following way: if $\mathbf{v} \in V$ and $A \in X$ then there is an element $\mathbf{v}(A) \in X$, and

1. If $\mathbf{u}, \mathbf{v} \in V$ and $A \in X$, then $(\mathbf{u} + \mathbf{v})(A) = \mathbf{u}(\mathbf{v}(A))$, and
2. For any two points $A, B \in X$ there exists a unique vector $\mathbf{v} \in V$ such that $\mathbf{v}(A) = B$. This vector $\mathbf{v}$ is denoted $\overrightarrow{AB}$.

The elements of X are called the **points** of the affine space (X, V) , and sometimes, when no confusion will result, one speaks of “the affine space X .”

A **real affine space** is an affine space (X, V) where V is a vector space over the field $\mathbb{R}$ of real numbers.

If (X, V) is an affine n -space, W is a (vector) subspace of V of dimension $m \leq n$, and A is a fixed point in X , then the **affine subspace determined by A and W** , denoted $S(A, W)$, is the set of all points $\mathbf{w}(A)$ as $\mathbf{w}$ ranges over W . A subset of X is called an affine subspace of dimension m if it is of the form $S(A, W)$ for some A in X and subspace W of dimension m (cf. Fact 2, below).

A one-dimensional affine subspace of X is called a **line**; a two-dimensional affine subspace, a **plane**.

The vector subspace W is called the **direction space** of the affine subspace $S(A, W)$.

Two affine subspaces of the same dimension are **parallel** if they have the same direction space. More generally, two affine subspaces (of possibly different dimension) are **parallel** if the direction space of one is a subspace of the direction space of another.

If C is a point in X , the map $P \rightarrow \overrightarrow{CP}$ is a bijection between X and V . By identifying a point of X with its image in V under this bijection, addition and scalar multiplication in X can be defined so that it becomes a vector space isomorphic to V , called the **tangent space** at C and denoted $X(C)$.

The elements of a set $\{X_1, \dots, X_d\}$ of d points in an affine space are **in general position**, or **affine-independent**, if they are not contained in any affine subspace of dimension less than or equal to $d - 2$.

Three points that are affine-independent are called **noncollinear**.

If A and B are distinct points in a real affine space, the set of points P such that $\overrightarrow{AP} = t\overrightarrow{AB}$ for some t , $0 \leq t \leq 1$, is called the **line segment** from A to B and is denoted $[A, B]$. The point P which corresponds to $t = 1/2$ is the **midpoint** of the line segment.

If A, B , and C are three points in general position in a real affine space, then the set of points P such that $\overrightarrow{AP} = t\overrightarrow{AB} + u\overrightarrow{AC}$, where u and t are nonnegative real numbers whose sum does not exceed 1, is called the **triangle** with vertices A, B , and C .

A subset Y of a real affine space is **convex** if whenever A and B are in Y , every point on the line segment $[A, B]$ is also in Y .

If (X, V) is an affine n -space and $n \geq 2$, then a bijection $T: X \rightarrow X$ is **semiaffine** if the image under T of any d -dimensional affine subspace is also a d -dimensional affine subspace, and is **affine** if it is semiaffine and satisfies the following additional condition: If A and B are distinct points and C and D are distinct points, and $\overrightarrow{AB} = k\overrightarrow{CD}$ for some nonzero scalar k , then $\overrightarrow{T(A)T(B)} = k\overrightarrow{T(C)T(D)}$.

If (X, V) is an affine n -space and $\mathbf{v} \in V$, the translation $T_{\mathbf{v}}$ is the map $T_{\mathbf{v}}: X \rightarrow X$ defined by $T_{\mathbf{v}}(A) = \mathbf{v}(A)$.

Facts: For proofs, see [ST71].

1. If A and B are points in an affine space, then
 - (a) $\overrightarrow{AB} = 0$ if and only if $A = B$.
 - (b) $\overrightarrow{AB} = -\overrightarrow{BA}$.
 - (c) $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$.
2. If $S = S(A, W)$ is an affine subspace of the affine space (X, V) , then $W = \{\overrightarrow{AB}: A, B \in S\}$. In particular, W is uniquely determined by S .
3. The affine subspaces $S(A, W)$ and $S(B, W)$ are equal if and only if $\overrightarrow{AB} \in W$.
4. The intersection of two affine subspaces is either empty or is an affine subspace. Specifically, if the point P is in both $S(A, U)$ and $S(B, W)$, then $S(A, U) \cap S(B, W) = S(P, U \cap W)$.

5. (Generalized Euclidean Parallel Postulate) If $S = S(A, W)$ is a d -dimensional affine subspace of an affine space X , and P is any point of X , then there exists a unique d -dimensional affine subspace of X , namely $S(P, W)$, that is parallel to S and contains the point P .
6. Given any two distinct points A and B in an affine space, there is a unique line containing them both.
7. The intersection of two convex subsets of a real affine space is convex (possibly empty). In addition, every affine subspace of a real affine space is convex.
8. The set of all semiaffine transformations of an affine n -space (X, V) forms a group under the operation of function composition. The set of all affine transformations of the affine space is a subgroup of this group.
9. A semiaffine transformation of an affine space of dimension at least two maps parallel subspaces to parallel subspaces, i.e., if S and S' are parallel subspaces of the affine space X and T is a semiaffine transformation, then $T(S)$ and $T(S')$ are parallel.
10. If (X, V) is an affine space of dimension at least two and V is a vector space over a field that, like the field of real numbers, has no nontrivial automorphisms, then any semiaffine transformation is an affine transformation.
11. A translation of an affine n -space, where n is at least two, is an affine transformation of that space.
12. If $\{A_0, \dots, A_n\}$ and $\{B_0, \dots, B_n\}$ are two sets of points in general position in an affine n -space X , then there is a unique affine transformation T such that $T(A_i) = B_i$ for each $i = 0, 1, \dots, n$.
13. If C is an arbitrary but fixed point of X and T is an affine transformation of X , then T can be realized as the composition of a mapping that is a nonsingular linear transformation on the tangent space $X(C)$, followed by a translation of X .

Examples:

1. If V is any vector space over a field F , then by taking $X = V$ and $\mathbf{v}(A) = \mathbf{v} + A$ we obtain an affine space, denoted $\mathcal{A}(V)$. For two points (i.e., vectors) A and B , $\overrightarrow{AB} = B - A$. The affine subspaces of $\mathcal{A}(V)$ are precisely the additive cosets of the vector subspaces. When $V = F^n$, the vector space of column n -tuples of elements of a field F , the resulting affine space is often denoted $A^n(F)$.
2. Any affine subspace $S = \mathbf{v} + W$ of the affine space $\mathcal{A}(V)$ is an affine space (S, W) , with $\mathbf{u}(\mathbf{v} + W) = (\mathbf{u} + \mathbf{v}) + W$. It can also be an affine space with a different (but isomorphic) vector space, however. For example, in the affine space $A^2(F)$, let S be the set of all points $\begin{bmatrix} x \\ y \end{bmatrix}$ which satisfy $x + y = 1$, so (S, W) is an affine space with $W = \left\{ \begin{bmatrix} u \\ -u \end{bmatrix} : u \in F \right\}$. But S is also an affine space with associated vector space F , with the action of F on W given by $u \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + u \\ y - u \end{bmatrix}$ for $u \in F$.
3. In the affine space $\mathcal{A}(V)$, the tangent space $V(\mathbf{0})$ is identical, not just as a set, but also with regard to the vector space operations, to the vector space V . Fact 13 therefore implies that any affine transformation of this affine space is of the form $T(\mathbf{v}) = L(\mathbf{v}) + \mathbf{b}$, where L is a nonsingular linear transformation from V to itself and $\mathbf{b}$ is a fixed vector.
4. The vectors $\{v_0, \dots, v_n\}$ in $\mathcal{A}(V)$ are affine-independent if and only if the vectors $\{v_1 - v_0, \dots, v_n - v_0\}$ are linearly independent in the vector space V . Thus, for example, if $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ and $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ are two sets of vectors in general position in the affine space $\mathcal{A}(V)$, where V is two-dimensional, then to find the unique affine transformation mapping $\mathbf{u}$ to $\mathbf{x}$, $\mathbf{v}$ to $\mathbf{y}$, and $\mathbf{w}$ to $\mathbf{z}$, we first use translation by $-\mathbf{u}$ to map $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ to $\{\mathbf{0}, \mathbf{v} - \mathbf{u}, \mathbf{w} - \mathbf{u}\}$, then use the (unique) linear transformation that maps the linearly independent vectors $\mathbf{v} - \mathbf{u}$ and $\mathbf{w} - \mathbf{u}$ to $\mathbf{y} - \mathbf{x}$ and $\mathbf{z} - \mathbf{x}$, respectively, and finally use translation by $\mathbf{x}$.

5. As a specific illustration of the idea discussed in the previous example, consider $A^2(\mathbb{R})$. Let $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$, and $\mathbf{z} = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$. If T denotes translation by $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$, then T maps $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, respectively, to $\mathbf{u}' = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mathbf{v}' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and $\mathbf{w}' = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The linear transformation L that maps $\mathbf{v}'$ and $\mathbf{w}'$ to $\mathbf{y}-\mathbf{x}$ and $\mathbf{z}-\mathbf{x}$ is given by the matrix $\begin{bmatrix} 3 & 5 \\ 3 & 8 \end{bmatrix}$. Thus, the composite transformation LT , which in coordinates is given by $\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} 3x + 5y - 8 \\ 3x + 8y - 11 \end{bmatrix}$, maps $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ to $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 5 \\ 8 \end{bmatrix}$. Composing this composite map with translation by $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ gives an affine map which maps $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ to $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$. In coordinates, this affine map is given by $\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} 3x + 5y - 7 \\ 3x + 8y - 7 \end{bmatrix}$, which can be viewed as $T'L$, where T' is translation by $\begin{bmatrix} -7 \\ -7 \end{bmatrix}$.
6. If $\{2\mathbf{u}, 2\mathbf{v}, 2\mathbf{w}\}$ is a set of three vectors in general position in the affine plane $A^2(\mathbb{R})$, then we can think of these vectors as the vertices of a triangle in the Cartesian plane. (The coefficients appear simply to make subsequent calculations less messy.) The midpoint of the line containing $2\mathbf{u}$ and $2\mathbf{v}$ is $\mathbf{u} + \mathbf{v}$, and similarly for the other sides of the triangle. Thus, the median of the triangle emanating from vertex $2\mathbf{w}$ is the line containing $2\mathbf{w}$ and $\mathbf{u} + \mathbf{v}$, which is the coset $2\mathbf{w} + \text{Span}(\mathbf{u} + \mathbf{v} - 2\mathbf{w})$. It is easy to verify that $(1/3)(2\mathbf{u} + 2\mathbf{v} + 2\mathbf{w})$ is on this line. Similar calculations show that this point is on the other two medians as well. This gives an algebraic proof of the familiar result from high school geometry that the medians of a triangle are concurrent.
7. Let F be a field that has a nonidentity automorphism f , such as the complex numbers with complex conjugation. In the affine plane $A^2(F)$, the mapping $T: \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} f(x) \\ f(y) \end{bmatrix}$ is a semiaffine transformation that is not affine. To verify that it is not affine, let a be an element of F that is not fixed by f , and consider the vectors $\mathbf{u} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ a \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, and $\mathbf{z} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. It is then easily verified that $\overrightarrow{\mathbf{u}\mathbf{v}} = a(\overrightarrow{\mathbf{w}\mathbf{z}})$ but $\overrightarrow{T(\mathbf{u})T(\mathbf{v})} = f(a)\overrightarrow{T(\mathbf{w})T(\mathbf{z})}$.

65.2 Euclidean Spaces

Definitions:

Euclidean n -space is an affine n -space (X, V) where V is a real inner product space. When $n = 2$, Euclidean n -space is called a **Euclidean plane**.

The **distance** between two points A and B in the Euclidean n -space (X, V) is $||\overrightarrow{AB}||$, where the norm is taken pursuant to the inner product in V . This distance is denoted $d(A, B)$.

Two lines in a Euclidean plane are **orthogonal** if the vectors that span their respective direction spaces are orthogonal with respect to the inner product of V .

If A and B are distinct points in a Euclidean plane, the **perpendicular bisector** of line segment $[A, B]$ is the line passing through the midpoint of this segment whose direction space is the orthogonal complement of the space spanned by the vector $\overrightarrow{AB}$.

An **isometry** or **rigid motion** of a Euclidean n -space (X, V) is a bijection T of X that preserves distances: $d(A, B) = d(T(A), T(B))$, for all A and B in X .

The **linear transformation associated with** T is the mapping $T': V \rightarrow V$ defined by $T'(\overrightarrow{AB}) = \overrightarrow{T(A)T(B)}$ (cf. Fact 4, below).

The isometry T is **direct** if T' has positive determinant, **indirect** if T' has negative determinant.

The remaining definitions in this section apply to the Euclidean plane $A^2(\mathbb{R})$ with the ordinary dot product as the inner product. (Recall from the preceding section that this is the affine plane obtained by taking $X = V = \mathbb{R}^2$.)

A 2×2 **rotation matrix** is a matrix of the form $R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.

A 2×2 **reflection matrix** is a matrix of the form $S_\theta = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$.

If C is a fixed point in $\mathbb{R}^2$, a **rotation about point C** is a mapping from $\mathbb{R}^2$ to itself of the form $T(X) = R_\theta(X - C) + C$, for some 2×2 rotation matrix R_θ .

If l is a line, the **reflection through l** is the mapping from $A^2(\mathbb{R})$ to itself which sends every point on l to itself, and which sends a point C not on l to the unique point C' with the property that l is the perpendicular bisector of the line segment $[C, C']$.

A **glide reflection** is a reflection through a line l followed by translation by a nonzero vector that spans the direction space of l .

Facts: For proofs, see [Roe93] and [Ree83].

1. The distance function d on a Euclidean n -space is a metric on the set X . Specifically, this means that if A , B , and C are arbitrary points of X , then
 - (a) $d(A, B)$ is nonnegative, and equal to 0 if and only if $A = B$.
 - (b) $d(A, B) = d(B, A)$.
 - (c) $d(A, B) \leq d(A, C) + d(C, B)$.
2. The point C is in the line segment $[A, B]$ if and only if $d(A, C) + d(C, B) = d(A, B)$.
3. (Pythagorean Theorem) If A , B , and C are three noncollinear points in a Euclidean n -space and $\overrightarrow{AB}$ is orthogonal to $\overrightarrow{BC}$ then $d(A, B)^2 + d(B, C)^2 = d(A, C)^2$.
4. An isometry is an affine mapping.
5. The associated linear transformation of an isometry is a well-defined orthogonal linear transformation.
6. In the Euclidean space $A^n(\mathbb{R})$, when an isometry is written as $T(\mathbf{v}) = \mathbf{L}(\mathbf{v}) + \mathbf{b}$ for a nonsingular linear transformation L and vector $\mathbf{b}$, the associated linear transformation T' is simply L . (Such an expression for an isometry T is always possible.)
7. Any translation in a Euclidean n -space is an isometry of that space. The associated linear transformation of a translation is the identity map.
8. The set of all isometries of a Euclidean space (X, V) is a group (under the operation of function composition).
9. The map that associates to every isometry T its associated linear transformation T' is a homomorphism from the isometry group onto the group of all orthogonal linear transformations of V , the kernel of which is the set of all translations.
10. If l is a line in the affine space $A^2(\mathbb{R})$, then reflection through l is an isometry.
11. If l is a line that passes through the origin in $A^2(\mathbb{R})$, then reflection through l is a linear transformation given by a reflection matrix. Conversely, the linear transformation defined by multiplication of a vector by a (fixed) reflection matrix S_θ is a reflection through the line passing through the origin with direction space spanned by $\begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{bmatrix}$.
12. A rotation is an isometry.
13. In $A^2(\mathbb{R})$, a rotation about the origin $\mathbf{0}$ is a linear transformation given by a rotation matrix.
14. In $A^2(\mathbb{R})$, the point C is the unique fixed point of a nonidentity rotation about C . Any isometry with a unique fixed point is a rotation.
15. Any isometry of $A^2(\mathbb{R})$ can be written as the product of three or fewer reflections.

16. Every isometry of $A^2(\mathbb{R})$ is either a translation, rotation, reflection, or glide reflection.
17. In $A^2(\mathbb{R})$, the product of a reflection with itself is the identity mapping. The product of two distinct reflections is a translation if the reflections are through parallel lines; if the reflections are through two lines intersecting at a point, the product of the two reflections is a rotation about that point. The product of three reflections is either a reflection or glide reflection.
18. The translations and rotations are the direct isometries of $A^2(\mathbb{R})$ and the reflections and glide reflections are the indirect ones.
19. Two triangles ABC and $A'B'C'$ in $A^2(\mathbb{R})$ are congruent in the sense of high school geometry (i.e., corresponding sides and angles are equal) if and only if there is an isometry of $A^2(\mathbb{R})$, which maps A to A' , B to B' , and C to C' .

Examples:

1. In $A^2(\mathbb{R})$, the line segment between $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ lies on a line with direction space spanned by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$; the span of $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ is the orthogonal complement of this direction space. The midpoint of this line segment is $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$. Therefore, the perpendicular bisector of this segment is the line l given by $\begin{bmatrix} 2 \\ 2 \end{bmatrix} + \text{Span} \left(\begin{bmatrix} -1 \\ 1 \end{bmatrix} \right)$.
2. In $A^2(\mathbb{R})$ the map $\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} -y \\ -x \end{bmatrix}$ has matrix $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$, which is a reflection matrix. To determine the line through which this mapping is a reflection, note that its eigenvalues are 1 and -1 . An eigenvector for the eigenvalue 1 is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Thus, the line spanned by $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is the line through which this mapping reflects. Alternatively, $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = S_{3\pi/2}$, so by Fact 11, the line of reflection is spanned by $\begin{bmatrix} \cos(3\pi/4) \\ \sin(3\pi/4) \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.
3. The mapping $\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} -y \\ x \end{bmatrix}$ has matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, which is a rotation matrix (corresponding to $\theta = \pi/2$). The mapping $T: \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} -y+1 \\ x \end{bmatrix}$, which is a direct isometry and not a translation, must correspond to a rotation also, through a point other than the origin. To determine the center of this rotation, we compute the unique fixed point to be $x = 1/2 = y$. Thus, T is a rotation around the point $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$.

65.3 Projective Spaces

Definitions:

If V is an $(n+1)$ -dimensional vector space over a field F , then the n -dimensional projective space based on V , denoted $\mathcal{P}(V)$, is the set of all subspaces of V . The one-dimensional subspaces of V are the **points** of $\mathcal{P}(V)$; the two-dimensional subspaces of V are the **lines** of $\mathcal{P}(V)$. More generally, the k -dimensional projective subspaces are the $(k+1)$ -dimensional subspaces of V .

When V has dimension 3, $\mathcal{P}(V)$ is called the **projective plane** based on V .

A point $\text{Span}(\mathbf{v})$ **lies on** a projective subspace if it is a subset of that projective subspace.

Relative to a fixed ordered basis of V , any nonzero element of V can be identified with an $(n+1)$ -tuple of elements of F . Thus, by selecting a spanning vector of any point of $\mathcal{P}(V)$, that point can be identified with an $(n+1)$ -tuple of elements of F , where not all of the components are zero and where two such $(n+1)$ -tuples are identified if one is a nonzero scalar multiple of the other. Under this identification, the $(n+1)$ -tuple is denoted $[a_1 : a_2 : \dots : a_{n+1}]$ and called the **homogeneous coordinates** of the point. (In many geometry books, homogenous coordinates are denoted $[a_1, a_2, \dots, a_{n+1}]$, but here that notation risks confusion with a $1 \times n$ matrix.)

When $\mathcal{P}(V)$ is a projective space of dimension n , with ordered basis $\mathcal{B}$ for V , then for any r -dimensional subspace W of V there is an $(n+1-r)$ -dimensional subspace of the dual space V^* consisting of those linear functionals of V that vanish on W (cf. Section 3.8). In particular, if $\mathcal{P}(V)$ is a projective plane, then points (respectively, lines) of $\mathcal{P}(V)$ correspond to lines (respectively, points) of $\mathcal{P}(V^*)$. Since, relative to the dual basis of V^* , any point can be given homogenous coordinates, any line of $\mathcal{P}(V)$ can be given the homogenous coordinates of its annihilator; these are **homogenous line coordinates**.

The **dual** of a statement concerning points and lines in a projective plane is the statement obtained by interchanging the terms “point” and “line.”

If $\{A, B, C\}$ and $\{A', B', C'\}$ are two sets of noncollinear points in a projective plane $\mathcal{P}(V)$, then the triangles ABC and $A'B'C'$ are **perspective from the point** P if and only if the lines containing A and A' , B and B' , and C and C' all pass through the point P .

The triangles ABC and $A'B'C'$ are **perspective from the line** l if the points of intersection of the corresponding sides of the triangles all lie on the line l .

If T is a nonsingular linear transformation of V , then the mapping $\hat{T}$ from $\mathcal{P}(V)$ to itself, defined by mapping $\text{Span}(\{\mathbf{v}_1, \dots, \mathbf{v}_m\})$ to $\text{Span}\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_m)\}$, is called the **projective transformation** (or **projectivity**) **determined by** T .

A **collineation** of $\mathcal{P}(V)$ is a bijective mapping from $\mathcal{P}(V)$ onto itself which maps subspaces to subspaces of the same dimension and which preserves set inclusion.

Facts: For proofs see [Kap69] or specific references.

1. In a projective plane $\mathcal{P}(V)$, points and lines satisfy the following incidence properties:
 - (a) Two distinct points A and B lie on a unique line, denoted AB .
 - (b) Two distinct lines meet in a unique point.
 - (c) There are four distinct points, no three of which are collinear.
2. In the projective plane $\mathcal{P}(V)$, the dual of any theorem involving the incidence of points and lines is also a theorem. (In the preceding Fact, for example, statement (b) is the dual of statement (a), and vice versa.)
3. [Art57] Any projective transformation of the projective space $\mathcal{P}(V)$ is a collineation of $\mathcal{P}(V)$. If F is a field that does not have any nontrivial automorphisms (such as, for example, the field of real numbers), then any collineation of the projective space $\mathcal{P}(V)$ is a projective transformation.
4. In the projective plane $\mathcal{P}(V)$, if a point P has homogenous coordinates $[a:b:c]$ and a line l has homogenous coordinates $[x:y:z]$, then P lies on l if and only if $ax + by + cz = 0$. Thus, to find the homogenous line coordinates of the line containing the points $[a:b:c]$ and $[d:e:f]$, we can use the

formula for vector cross product to find a vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, which is orthogonal to the two vectors $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$

and $\begin{bmatrix} d \\ e \\ f \end{bmatrix}$ under the usual dot product. For an arbitrary field F , this formula yields the homogenous line coordinates $[x:y:z]$ of the line containing the two points.

5. (Desargues' Theorem and its converse) Two triangles in a projective plane $\mathcal{P}(V)$ are perspective from a point if and only if they are perspective from a line.
6. (Pappus' Theorem) Let A, B, C and A', B', C' be two triples of distinct collinear points in a projective plane. Let P, Q , and R denote, respectively, the points of intersection of the lines AB' and $A'B$, AC' and $C'A$, and BC' and $B'C$. Then P, Q , and R are collinear.
7. Let A, B, C , and D be four distinct points in a projective plane $\mathcal{P}(V)$ with the property that no three are collinear. Let A', B', C' , and D' be four other distinct points with this property. Then there is a unique projective transformation which maps A to A' , B to B' , C to C' , and D to D' .

Examples:

1. If V is a two-dimensional vector space over the field of complex numbers, then the projective line $\mathcal{P}(V)$ can be realized as the set of all homogeneous coordinates $[x: y]$ where x and y are complex numbers, not both zero. By identifying $[x: y]$ with the complex number x/y if y is nonzero, and with ∞ if $y = 0$, we can identify $\mathcal{P}(V)$ with the extended complex plane studied in courses on complex analysis. Note also that the complex-linear transformation of V given by a nonsingular matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ represents the mapping of the extended plane given by $z \mapsto \frac{az + b}{cz + d}$; these are the linear fractional transformations studied in such a course.
2. The smallest projective space of dimension greater than 1 is obtained by letting V be a three-dimensional vector space over the two-element field Z_2 . In homogenous coordinates, every element of $\mathcal{P}(V)$ is represented by a triple $[a: b: c]$ where each entry is either 0 or 1 and not all entries are 0. Thus, there are seven points in this projective plane, and by duality there are seven lines as well. It is easy to verify that each line of this projective plane consists of three points, and each point is contained on three lines. For example, the line containing the points $[1:1:1]$ and $[1:0:0]$, obtained by taking the span of these vectors, contains these points, the point with homogenous coordinates $[0:1:1]$, and no other points (i.e., no other nonzero vector spans a one-dimensional subspace of the span of these two points).
3. If V is a three-dimensional vector space over the field of real numbers, then, thinking of the points of $\mathcal{P}(V)$ in terms of homogenous coordinates relative to a given fixed ordered basis of V , the points $[0:1:0]$ and $[0:0:1]$ lie on a unique line, which is the span of these vectors (i.e., the two-dimensional subspace of V consisting of all points with homogenous coordinates with first component zero). In the dual space of V , relative to the dual basis of the given ordered basis of V , this subspace corresponds to the subspace of all linear functionals of V that annihilate all such vectors. This is clearly the span of the first vector in the ordered dual basis. Thus, the homogenous line coordinates for the line containing the two given points is $[1:0:0]$. This can also be obtained as a cross product.
4. To illustrate Pappus' Theorem, let six points be given as follows in the real projective plane: $A = [1: 0: 0]$, $B = [0: 1: 0]$, $C = [1: 1: 0]$, $A' = [1: 1: 2]$, $B' = [1: 2: 2]$, $C' = [0: 0: 1]$. It was observed above that the line BC' consists of all points with homogenous coordinates with first component zero. It is easy to see that the line containing C and B' is the set of triples of real numbers of the form $\begin{bmatrix} a+b \\ a+b \\ a \end{bmatrix}$. The intersection of these two lines is obtained by letting $a = -b$, and, therefore, is the point with homogenous coordinates $[0:0:1]$. By similar calculations, the intersection of the lines AB' and $A'B$ is seen to be the point with homogenous coordinates $[1:2:2]$ and the intersection of the lines CA' and $C'A$ is the point with homogenous coordinates $[0:0:1]$. Since the vectors $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are linearly dependent, the one-dimensional subspaces spanned by these vectors are collinear points, as required by Pappus' Theorem.

5. The three incidence relations described in Fact 1 are usually taken as the defining relations for an axiomatic definition of projective plane, in which “point,” “line,” and “incidence” are taken as undefined notions. It is not the case that any projective plane defined axiomatically in this way is of the form $\mathcal{P}(V)$ for some vector space V over a field F . As an example, if we interpret “point” to mean the set of all ordered pairs of real numbers and “line” to mean any horizontal Euclidean line, vertical Euclidean line, Euclidean line of negative slope, or broken Euclidean line of positive slope m above the x -axis and slope $2m$ below the x -axis, we obtain a geometric system that satisfies the standard incidence relations for points and lines, and the Euclidean parallel postulate. We can convert this example into a projective plane by the addition of “ideal points”: To every line we add a new point, with the same point being added to two lines if and only if they are parallel. We also add a new line consisting of the ideal points, and no other points. The augmented geometric system satisfies the axioms of a projective plane, but neither Desargues’ Theorem nor Pappus’ Theorem holds in this plane. (This example is called the *Moulton Plane*.)
6. The addition of ideal points, as in the preceding example, can be carried out in any affine plane $A^2(F)$. An affine point $\begin{bmatrix} a \\ b \end{bmatrix}$ can be identified with the projective point with homogenous coordinates $[a:b:1]$ and the ideal points are those with homogenous coordinates with a zero in the third component.
7. In a projective n -space $\mathcal{P}(V)$, let H be a subspace of V of dimension n . Starting from an ordered basis for H , obtain an ordered basis for V by adding a single vector as the last element of the basis. Then, in homogenous coordinates, every point of $\mathcal{P}(V)$ that is not in H has nonzero last component. By associating the point with homogenous coordinates $[a_1 : \dots : a_{n+1}]$ with the n -tuple $\begin{bmatrix} a_1 \\ a_{n+1} \\ \vdots \\ a_n \\ a_{n+1} \end{bmatrix}$ we can think of the set of points that are not in H as the affine n -space $A^n(F)$.
8. Let $\mathcal{P}(V)$ be a projective plane, where V is a vector space over a field F that has a nonidentity automorphism f . The mapping $[a : b : c] \mapsto [f(a) : f(b) : f(c)]$ is a collineation. Since this mapping fixes the points $[0:0:1]$, $[0:1:0]$, $[1:0:0]$, and $[1:1:1]$ and is not the identity, it cannot, by Fact 7, be a projective transformation.
9. In the real projective plane, consider the six points $A = [1:2:1]$, $B = [2:0:1]$, $C = [5:5:1]$, $A' = [2:4:1]$, $B' = [4:0:1]$, and $C' = [10:10:1]$. Identifying the projective point with homogenous coordinates $[a:b:1]$ with the ordinary Euclidean point $\begin{bmatrix} a \\ b \end{bmatrix}$, as in Example 6 above, it can be seen by a simple diagram that the triangles ABC and $A'B'C'$ are perspective from the origin, since all lines AA' , BB' , and CC' pass through this point. Thus, in the real projective plane these triangles are perspective from the point $[0:0:1]$. Desargues’ Theorem, therefore, asserts that the points of intersection of the lines AB and $A'B'$, AC and $A'C'$, BC and $B'C'$ will lie on a line. A simple calculation shows that these three points, in homogenous coordinates, have third component zero, so these points are indeed collinear. This corresponds to the fact that as Euclidean lines there are no points of intersection — the corresponding sides of the triangles are parallel in pairs. The points with third component zero are the “ideal points” that have been added to Euclidean geometry to form the real projective plane.

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66

Some Applications of Matrices and Graphs in Euclidean Geometry

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This chapter presents some facts and examples illustrating the interplay between matrix theory, graph theory, and n -dimensional Euclidean geometry. The main objects of Euclidean geometry are, of course, the points. The simplest way of introducing them is to identify points with the endpoints of vectors; formally, one introduces an artificial point, the origin $\mathbf{O}$, and the points are sums of this origin with any vector. A more general treatment of Euclidean n -space can be found in Chapter 65.

66.1 Euclidean Point Space

Definitions:

An **arithmetic Euclidean vector space** is the real inner product space $\mathbb{R}^n$ with the standard inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^T \mathbf{x}$.

The **arithmetic point Euclidean n -space** E^n based on the vector space $\mathbb{R}^n$ has as **points** the column $n + 1$ -tuples with last coordinate 1, e.g., $\mathbf{C} = [c_1, c_2, \dots, c_n, 1]^T$, and as **vectors** the column $n + 1$ -tuples with last coordinate 0, e.g., $\mathbf{v} = [v_1, v_2, \dots, v_n, 0]^T$.

The origin $\mathbf{O}$ is the point $[0, \dots, 0, 1]^T$; the numbers $c_1, \dots, c_n$ are **coordinates** of the point $\mathbf{C}$.

Algebraic operations are defined with points: If $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m$ are points, $a_1, a_2, \dots, a_m$ real numbers, then the symbol

$$a_1 \mathbf{A}_1 + a_2 \mathbf{A}_2 + \dots + a_m \mathbf{A}_m$$

means a **point** if and only if $\sum a_k = 1$, and a **vector** if and only if $\sum a_k = 0$. In no other case is this symbol defined.

The **Euclidean distance**, or **distance**, of two points $\mathbf{A}$ and $\mathbf{B}$ is the **length** of the vector $\mathbf{A} - \mathbf{B}$, i.e., for $\mathbf{A} = [a_1, a_2, \dots, a_n, 1]^T$ and $\mathbf{B} = [b_1, b_2, \dots, b_n, 1]^T$,

$$\|\mathbf{A} - \mathbf{B}\| = \sqrt{(a_1 - b_1)^2 + \dots + (a_n - b_n)^2}.$$

Let $S = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m\}$ be a set of points in E^n . The point

$$\mathbf{C} = a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \dots + a_m\mathbf{A}_m \quad \text{and} \quad \sum_{k=1}^m a_k = 1,$$

is a **linear combination** of S . The set of all linear combinations of S is the **linear hull** of S , denoted by $\mathcal{L}(S)$.

A point $\mathbf{C}$ is **linearly dependent** on S if $\mathbf{C} \in \mathcal{L}(S)$. The set S is called **linearly dependent** if there is a point in S that is linearly dependent on the remaining points. Otherwise, the set S is called **linearly independent**.

Linear hulls of systems of points are called **linear subspaces** of E^n .

A linear subspace M has **dimension** k if the maximum number of linearly independent points in M is $k + 1$.

Linear subspaces of dimension 1 in E^n are called **lines**.

Linear subspaces of dimension $n - 1$ in E^n (i.e., of **codimension** 1) are called **hyperplanes**.

If $\hat{\alpha}$ is a hyperplane in E^n defined by the equation $\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n + \alpha_0 = 0$ (cf. Fact 10), the vector $\mathbf{u} = [\alpha_1, \dots, \alpha_n, 0]^T$ is a **normal vector** to $\hat{\alpha}$.

The **distance** of point $\mathbf{C}$ to a hyperplane $\hat{\alpha}$ is the minimum of all distances of the point $\mathbf{C}$ from the points in the hyperplane $\hat{\alpha}$.

Two hyperplanes are called **parallel** if their normal vectors are proportional, i.e., one is a scalar multiple of the other. They are called **orthogonal** (or **perpendicular**) if their normal vectors are orthogonal.

Let $S = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m\}$ be a set of points in E^n . Then the set of all points of the form

$$a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \dots + a_m\mathbf{A}_m,$$

for which all the coefficients a_i are nonnegative and $\sum a_k = 1$, is called the **convex hull** of S ; we denote it by $\mathcal{C}(S)$.

The convex hull of two distinct points $\mathbf{A}, \mathbf{B}$ is called the **segment** and is denoted by $\overline{\mathbf{AB}}$.

The point $\frac{1}{2}\mathbf{A} + \frac{1}{2}\mathbf{B}$ is the **midpoint** of the segment $\overline{\mathbf{AB}}$.

A set $S \in E^n$ is **convex** if S has the property that with any two points $\mathbf{A}$ and $\mathbf{B}$ in S , all points of the segment $\overline{\mathbf{AB}}$ are in S .

A hyperplane $\hat{\alpha}$ with normal vector $[\alpha_1, \alpha_2, \dots, \alpha_n]$ determines two **open halfspaces**; one is the set of all points $\mathbf{X} = [x_1, x_2, \dots, x_n, 1]^T$ satisfying

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n + \alpha_0 > 0,$$

while the other is the set of all points satisfying the reverse inequality.

In addition, one can speak about **closed halfspace** if in the inequality, equality is also admitted.

Facts:

These facts follow from facts in Chapter 1 and Chapter 65.

1. Points and vectors in arithmetic Euclidean n -space satisfy the following:
 - (E1.) The sum of a point and a vector is a point.
 - (E2.) The difference of two points is a vector.
 - (E3.) $(\mathbf{C} + \mathbf{v}) + \mathbf{w} = \mathbf{C} + (\mathbf{v} + \mathbf{w})$, where $\mathbf{C}$ is a point and $\mathbf{v}, \mathbf{w}$ are vectors.

2. Arithmetic Euclidean n -space, with vectors acting on points by addition, is a Euclidean n -space as defined in Chapter 65.
3. A linear hull is an affine subspace as defined in Chapter 65.
4. The set $\mathcal{S} = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m\}$ is linearly independent if and only if $\lambda_1 = \lambda_2 = \dots = \lambda_m = 0$ are the only numbers $\lambda_1, \dots, \lambda_m$ for which the zero vector $\mathbf{0}$ satisfies $\mathbf{0} = \lambda_1 \mathbf{A}_1 + \lambda_2 \mathbf{A}_2 + \dots + \lambda_m \mathbf{A}_m$ and $\sum_{k=1}^m \lambda_k = 0$.
5. The set $\mathcal{S} = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m\}$ is linearly independent if and only if every point $\mathbf{P} \in \mathcal{L}(\mathcal{S})$ has a unique expression as

$$\mathbf{P} = \lambda_1 \mathbf{A}_1 + \lambda_2 \mathbf{A}_2 + \dots + \lambda_m \mathbf{A}_m \text{ and } \sum_{k=1}^m \lambda_k = 1.$$

6. A linearly independent set in E^n contains at most $n + 1$ points. A linearly independent set with $n + 1$ linearly independent points in E^n exists.
7. Any linearly independent set has the property that each of its nonempty subsets is linearly independent as well.
8. Let the set $\mathcal{S}$ consist of m points $\mathbf{A}, \mathbf{B}, \dots, \mathbf{G}$ defined by

$$\mathbf{A} = [a_1, a_2, \dots, a_n, 1]^T, \mathbf{B} = [b_1, b_2, \dots, b_n, 1]^T, \dots, \mathbf{G} = [g_1, g_2, \dots, g_n, 1]^T.$$

Then $\mathcal{S}$ is linearly independent if and only if the $m \times (n + 1)$ matrix

$$\begin{bmatrix} a_1 & a_2 & \dots & a_n & 1 \\ b_1 & b_2 & \dots & b_n & 1 \\ \dots & \dots & \dots & \dots & \dots \\ g_1 & g_2 & \dots & g_n & 1 \end{bmatrix}$$

has rank m .

9. Let $\mathbf{A} = [a_1, a_2, \dots, a_n, 1]^T$, $\mathbf{B} = [b_1, b_2, \dots, b_n, 1]^T$, ..., $\mathbf{F} = [f_1, f_2, \dots, f_n, 1]^T$ be n linearly independent points in E^n . Then the linear hull of these points consists of all points $\mathbf{X} = [x_1, x_2, \dots, x_n, 1]^T$ in E^n which satisfy the equation

$$\det \begin{bmatrix} x_1 & x_2 & \dots & x_n & 1 \\ a_1 & a_2 & \dots & a_n & 1 \\ b_1 & b_2 & \dots & b_n & 1 \\ \dots & \dots & \dots & \dots & \dots \\ f_1 & f_2 & \dots & f_n & 1 \end{bmatrix} = 0.$$

10. A hyperplane can be characterized as the set of points $\mathbf{X} = [x_1, x_2, \dots, x_n, 1]^T$ such that the coordinates satisfy one linear equation of the form

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n + \alpha_0 = 0,$$

in which not all of the numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ are equal to zero.

11. We can generalize linear independence as well as the equation of the hyperplane in Fact 9 by including cases when some of the points (but not all of them) are replaced by vectors. We simply put into the corresponding row the $(n + 1)$ -tuple $a_1, a_2, \dots, a_n, 0$ instead that for the point.
12. Two parallel but distinct hyperplanes have no point in common. Conversely, if two hyperplanes in E^n , $n \geq 2$, have no point in common, then they are parallel.
13. Hyperplanes are parallel if and only if they are parallel as affine subspaces as defined in Chapter 65.
14. The definitions of segment and midpoint are equivalent to the definitions of these terms in Chapter 65 for arithmetic Euclidean n -space.

15. The distance of a point $\mathbf{C} = [c_1, c_2, \dots, c_n, 1]^T$ from a hyperplane $\hat{\alpha}$ given by equation $\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n + \alpha_0 = 0$ is

$$\frac{|\alpha_1 c_1 + \alpha_2 c_2 + \dots + \alpha_n c_n + \alpha_0|}{\sqrt{\alpha_1^2 + \alpha_2^2 + \dots + \alpha_n^2}}.$$

16. The distance of a point $\mathbf{C} = [c_1, c_2, \dots, c_n, 1]^T$ from a hyperplane $\hat{\alpha}$ given by equation $\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n + \alpha_0 = 0$ is the distance of $\mathbf{C}$ from the point $\mathbf{F} = \mathbf{C} - \gamma \mathbf{u}$, where $\mathbf{u} = [\alpha_1, \alpha_2, \dots, \alpha_n, 0]^T$ is the normal vector to $\hat{\alpha}$ and

$$\gamma = \frac{\alpha_1 c_1 + \alpha_2 c_2 + \dots + \alpha_n c_n + \alpha_0}{\alpha_1^2 + \alpha_2^2 + \dots + \alpha_n^2}.$$

Examples:

1. In E^3 , the points $\mathbf{A}_1 = [2, -1, 2, 1]^T$, $\mathbf{A}_2 = [1, 1, 3, 1]^T$, and $\mathbf{A}_3 = [0, 0, 1, 1]^T$ are linearly independent since the rank of the matrix

$$\begin{bmatrix} 2 & -1 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

is 3.

2. The point $\mathbf{A}_4 = [2, 0, 3, 1]^T$ is linearly dependent on the points $\mathbf{A}_1, \mathbf{A}_2$, and $\mathbf{A}_3$ from Example 1, since $\mathbf{A}_4 = \frac{2}{3}\mathbf{A}_1 + \frac{2}{3}\mathbf{A}_2 - \frac{1}{3}\mathbf{A}_3$.
3. The equation of the hyperplane (which is a plane) determined by the points in Example 1 is

$$0 = \det \begin{bmatrix} x_1 & x_2 & x_3 & 1 \\ 2 & -1 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} = -3x_1 - 3x_2 + 3x_3 - 3.$$

4. The triangle with vertices $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ from Example 1 is the convex hull of the three points $\mathbf{A}_1, \mathbf{A}_2$, and $\mathbf{A}_3$.
5. Let $n \geq 3$ be an integer. In the Euclidean n -space E^n , define points

$$\mathbf{F}_k = [\underbrace{n-k, n-k, \dots, n-k}_{k\text{-times}}, \underbrace{-k, -k, \dots, -k}_{(n-k)\text{-times}}, 1], \quad k = 1, \dots, n;$$

$$\mathbf{F}_0 = [n-1, n-2, \dots, 0, 1]^T, \text{ and}$$

$$\mathbf{C} = \left[\frac{1}{2}(n-1), \frac{1}{2}(n-3), \frac{1}{2}(n-5), \dots, -\frac{1}{2}(n-3), -\frac{1}{2}(n-1), 1 \right]^T.$$

Observe first that the points $\mathbf{C}, \mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_n$ are linearly dependent since $\mathbf{C} = \frac{1}{n}\mathbf{F}_1 + \frac{1}{n}\mathbf{F}_2 + \dots + \frac{1}{n}\mathbf{F}_n$. On the other hand, the points $\mathbf{F}_0, \mathbf{F}_1, \dots, \mathbf{F}_n$ form a linearly independent set by Fact 8 since the determinant (of an $(n+1) \times (n+1)$ matrix)

$$\det \begin{bmatrix} n-1 & n-2 & n-3 & n-4 & \dots & 0 & 1 \\ n-1 & -1 & -1 & -1 & \dots & -1 & 1 \\ n-2 & n-2 & -2 & -2 & \dots & -2 & 1 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 2 & 2 & 2 & 2 & \dots & -(n-2) & 1 \\ 1 & 1 & 1 & 1 & \dots & -(n-1) & 1 \\ 0 & 0 & 0 & 0 & \dots & 0 & 1 \end{bmatrix}$$

is different from zero: Subtracting the $\frac{1}{n}$ -multiple of the sum of the second till last row, from the first row, the first row is a $\frac{1}{2}(n-1)$ -multiple of the row of all ones. Factoring out this number $\frac{1}{2}(n-1)$ from the determinant, subtracting the resulting first row from the n th row, the 2-multiple of the first row from the $(n-1)$ st row, etc., till the $(n-1)$ -multiple of the first row from the second row, we obtain the determinant of an upper triangular matrix with all diagonal entries different from zero. The value of the determinant is then also easily determined.

The Euclidean distances between the points $\mathbf{F}_i$ and $\mathbf{F}_{i+1}$, $i = 1, \dots, n-1$, as well as between $\mathbf{F}_n$ and $\mathbf{F}_1$, are all equal, equal to $\sqrt{n(n-1)}$. The point $\mathbf{C}$ has same distances from all points $\mathbf{F}_i$, $i = 1, 2, \dots, n$, equal to $\sqrt{\frac{1}{12}(n-1)n(n+1)}$.

All the points $\mathbf{F}_i$, $i = 1, 2, \dots, n$, as well as the point $\mathbf{C}$, are points of the hyperplane H with equation $\sum_{i=1}^n x_i = 0$.

The vector $\mathbf{F}_0 - \mathbf{C}$ is the $\frac{1}{2}(n-1)$ -multiple of the vector $\mathbf{u} = [1, 1, \dots, 1]^T$ (which is throughout denoted as $\mathbf{1}$). This vector is at the same time the normal vector to the hyperplane H . It follows that the distance of the point $\mathbf{F}_0$ from H is equal to the length of the vector $\mathbf{F}_0 - \mathbf{C}$, which is $\frac{1}{2}(n-1)\sqrt{n}$. The same result can be obtained using Fact 15.

The hyperplane with equation

$$x_1 + x_2 + \dots + x_n - 1 = 0$$

is parallel to H ; the hyperplane

$$x_1 - x_2 = 0$$

is orthogonal to H since the vector $[1, -1, 0, \dots, 0, 0]^T$ is orthogonal to the vector $\mathbf{u}$.

66.2 Gram Matrices

Definitions:

The **Gram matrix** $G(S)$ of an ordered system $S = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m)$ of vectors in the Euclidean vector n -space $\mathbb{R}^n$ is the $m \times m$ matrix $G(S) = G(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m) = [\langle \mathbf{a}_i, \mathbf{a}_j \rangle]$. (See also Section 8.1.).

Let $\mathbf{O}, \mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k$, $k \geq 2$, be linearly independent points in E^n , and let $\mathbf{u}_1 = \mathbf{A}_1 - \mathbf{O}$, $\mathbf{u}_2 = \mathbf{A}_2 - \mathbf{O}$, $\dots$, $\mathbf{u}_k = \mathbf{A}_k - \mathbf{O}$, be the corresponding vectors. We call the set of all points of the form $\mathbf{O} + \sum_{i=1}^k a_i \mathbf{u}_i$, where the numbers a_i satisfy $0 \leq a_i \leq 1$, $i = 1, \dots, k$, the **parallelepiped** spanned by the vectors $\mathbf{u}_i$. For $k = 2$, we speak about the **parallelogram** spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$.

If $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ is a basis of an n -dimensional arithmetic Euclidean vector space and $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a set of vectors such that the inner product of $\mathbf{u}_i$ and $\mathbf{v}_j$ is the Kronecker delta δ_{ij} , then this pair of ordered sets is a **biorthogonal pair of bases**.

Facts:

Facts for which no specific reference is given follow from facts in Chapter 1.

1. The Gram matrix is always a positive semidefinite matrix. Its rank is equal to the dimension of the Euclidean space of smallest dimension which contains all the vectors of the system.
2. Every positive semidefinite matrix is a Gram matrix of some system of vectors S in some Euclidean space.
3. Every linear relationship among the vectors in S is reflected in the same linear relationship among the rows of $G(S)$, and conversely.
4. The k -dimensional volume of the parallelepiped spanned by the vectors

$\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ is

$$\sqrt{\det G(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k)}.$$

5. To every basis $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ of an n -dimensional Euclidean vector space there exists a set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ such that this pair is a biorthogonal pair of bases. The set of the $\mathbf{v}_i$ s is also a basis and is uniquely determined.
6. If both bases in the biorthogonal pair coincide, the common basis is orthonormal, and an orthonormal basis forms a biorthogonal pair with itself.
7. The Gram matrices $G(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n)$ and $G(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ of a pair of biorthogonal bases are inverse to each other:

$$G(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n)G(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = I.$$

8. [Fie64] Let $A = [a_{ij}]$ be a positive semidefinite matrix with row sums zero. Then

$$2 \max_i \sqrt{a_{ii}} \leq \sum_i \sqrt{a_{ii}}.$$

9. [Fie61b] If A is positive definite, then the matrix $A \circ A^{-1} - I$ is positive semidefinite and its row sums are equal to zero.
10. If $A = [a_{ij}]$ is positive definite and $A^{-1} = [\alpha_{ij}]$, then

$$a_{ii}\alpha_{ii} \geq 1 \quad \text{for all } i$$

and

$$2 \max_i \sqrt{a_{ii}\alpha_{ii} - 1} \leq \sum_i \sqrt{a_{ii}\alpha_{ii} - 1}.$$

11. [Fie64] Let $A = [a_{ik}]$ be a positive definite matrix, and let $A^{-1} = [\alpha_{ik}]$. Then the diagonal entries of A and A^{-1} satisfy the first condition in Fact 10 and

$$2 \max_i (\sqrt{a_{ii}\alpha_{ii}} - 1) \leq \sum_i (\sqrt{a_{ii}\alpha_{ii}} - 1).$$

Conversely, if some n -tuples of positive numbers a_{ii} and α_{ii} satisfy these conditions, then there exists a positive definite $n \times n$ matrix A with diagonal entries a_{ii} such that the diagonal entries of A^{-1} are α_{ii} .

12. [Fie64] Let the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ form a pair of biorthogonal bases in a Euclidean n -space E . Then

$$\|\mathbf{u}_i\| \|\mathbf{v}_i\| \geq 1, \quad i = 1, \dots, n,$$

$$2 \max_i (\|\mathbf{u}_i\| \|\mathbf{v}_i\| - 1) \leq \sum_i (\|\mathbf{u}_i\| \|\mathbf{v}_i\| - 1).$$

Conversely, if nonnegative numbers $\alpha_1, \alpha_2, \dots, \alpha_n, \beta_1, \beta_2, \dots, \beta_n$ satisfy

$$\alpha_i \beta_i \geq 1, \quad i = 1, \dots, n,$$

$$2 \max_i (\alpha_i \beta_i - 1) \leq \sum_i (\alpha_i \beta_i - 1),$$

then there exists in E^n a pair of biorthogonal bases $\mathbf{u}_i, \mathbf{v}_j$, such that

$$\|\mathbf{u}_i\| = \alpha_i, \|\mathbf{v}_i\| = \beta_i, \quad i = 1, \dots, n.$$

13. [Fie64] Let $A = [a_{ij}]$ be an $n \times n$ positive definite matrix, $n \geq 2$, and let $A^{-1} = [\alpha_{ij}]$. Then the following are equivalent:

- (a) $\sqrt{a_{nn}\alpha_{nn}} - 1 = \sum_{i=1}^{n-1} (\sqrt{a_{ii}\alpha_{ii}} - 1).$
 (b) $\frac{a_{ij}}{\sqrt{a_{ii}}\sqrt{a_{jj}}} = \frac{\alpha_{ij}}{\sqrt{\alpha_{ii}}\sqrt{\alpha_{jj}}}, \quad i, j = 1, \dots, n-1, \text{ and}$
 $\frac{a_{in}}{\sqrt{a_{ii}}\sqrt{a_{nn}}} = -\frac{\alpha_{in}}{\sqrt{\alpha_{ii}}\sqrt{\alpha_{nn}}}, \quad i = 1, \dots, n-1.$
 (c) A is diagonally similar to

$$C = \begin{bmatrix} I_1 + \omega \mathbf{c}\mathbf{c}^T & \mathbf{c} \\ \mathbf{c}^T & 1 + \omega \mathbf{c}^T \mathbf{c} \end{bmatrix},$$

where $\mathbf{c}$ is a real vector with $n-1$ coordinates and

$$\omega = \frac{\sqrt{1 + \mathbf{c}^T \mathbf{c}} - 1}{\mathbf{c}^T \mathbf{c}}$$

if $\mathbf{c} \neq \mathbf{0}$; if $\mathbf{c} = \mathbf{0}$, $\omega = 0$.

14. To realize a positive definite $n \times n$ matrix C as a Gram matrix of some n vectors, say $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, it suffices to find a nonsingular matrix A such that $C = AA^T$ and to use the entries in the k th row of A as coordinates of the vector $\mathbf{a}_k$. Such matrix A can be found, e.g., by the Gram-Schmidt process (cf. Section 5.5).

Examples:

- The vectors $\mathbf{u}_1 = [1, 3]^T$, $\mathbf{u}_2 = [2, -1]^T$ in $\mathbb{R}^2$ are linearly independent and form a basis in $\mathbb{R}^2$.
The Gram matrix $G(\mathbf{u}_1, \mathbf{u}_2) = \begin{bmatrix} 10 & -1 \\ -1 & 5 \end{bmatrix}$ is nonsingular.
- To find the pair of vectors $\mathbf{v}_1, \mathbf{v}_2$ that form a biorthogonal pair with the vectors $\mathbf{u}_1, \mathbf{u}_2$ in Example 1, observe that $\mathbf{v}_1$ should satisfy $\langle \mathbf{u}_1, \mathbf{v}_1 \rangle = 1$ and $\langle \mathbf{u}_2, \mathbf{v}_1 \rangle = 0$. Set $\mathbf{v}_1 = [x_1, x_2]^T$; thus $x_1 + 3x_2 = 1$, $2x_1 - x_2 = 0$, i.e., $\mathbf{v}_1 = [\frac{1}{7}, \frac{2}{7}]^T$. Analogously, $\mathbf{v}_2 = [\frac{3}{7}, -\frac{1}{7}]^T$. The Gram matrix is $G(\mathbf{v}_1, \mathbf{v}_2) = \begin{bmatrix} \frac{5}{49} & \frac{1}{49} \\ \frac{1}{49} & \frac{10}{49} \end{bmatrix}$. It is easily verified that $G(\mathbf{v}_1, \mathbf{v}_2)$ is the inverse of $G(\mathbf{u}_1, \mathbf{u}_2)$, as stated in Fact 7.
- The pairs of vectors $(\mathbf{e}_1 = [1, 0]^T, \mathbf{e}_2 = [0, 1]^T)$ and $(\mathbf{u}_1 = [\frac{3}{5}, \frac{4}{5}]^T, \mathbf{u}_2 = [-\frac{4}{5}, \frac{3}{5}]^T)$ are orthonormal bases for $\mathbb{R}^2$, so $G(\mathbf{e}_1, \mathbf{e}_2) = I$ and $G(\mathbf{u}_1, \mathbf{u}_2) = I$ are inverses, but $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{u}_1, \mathbf{u}_2)$ are not a biorthogonal pair of bases.

66.3 General Theory of Euclidean Simplexes

An n -simplex in E^n is a generalization of the triangle in the plane and the tetrahedron in the three-dimensional space. Just as not every triplet of positive numbers can serve as lengths of three sides of a triangle (they have to satisfy the strict triangle inequality), we can ask about analogous conditions for the simplex.

Definitions:

An **n -simplex** is the convex hull of $n+1$ linearly independent points in E^n .

The points are called **vertices** of the simplex. The convex hull of a subset of the set of vertices is called a **face** of the simplex.

If the subset of vertices has $k+1$ elements, the face has **dimension** k . One-dimensional faces are called **edges**.

If $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{n+1}$ are the vertices of an n -simplex Σ , we write $\Sigma = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{n+1}\}$. The $(n-1)$ -dimensional face opposite $\mathbf{A}_i$ is denoted by ω_i . A vector $\mathbf{n}$ is an **outer normal** to ω_i if $\mathbf{n}$ is a normal to ω_i and for $\mathbf{C} \in \omega_i$, $\mathbf{C} + \mathbf{n}$ is not in the same halfspace as $\mathbf{A}_i$.

The **dihedral interior angle** between ω_i and ω_j , $i \neq j$ (opposite the edge $\mathbf{A}_i \mathbf{A}_j$) is $\pi -$ (the angle between an outer normal to ω_i and an outer normal to ω_j), and is denoted by ϕ_{ij} .

The **barycentric coordinates** with respect to the simplex $\Sigma = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{n+1}\}$ of a point $\mathbf{P}$ in $\mathcal{L}(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{n+1})$ are the unique numbers λ_i such that $\mathbf{P} = \lambda_1 \mathbf{A}_1 + \lambda_2 \mathbf{A}_2 + \dots + \lambda_{n+1} \mathbf{A}_{n+1}$ and $\sum_{k=1}^{n+1} \lambda_k = 1$ (cf. Fact 5 of section 66.1). Strictly speaking, these numbers are the **inhomogeneous barycentric coordinates**.

The **homogeneous barycentric coordinates** of $\lambda_1 \mathbf{A}_1 + \lambda_2 \mathbf{A}_2 + \dots + \lambda_{n+1} \mathbf{A}_{n+1}$ are the numbers λ_i (not all 0). The expression $\lambda_1 \mathbf{A}_1 + \lambda_2 \mathbf{A}_2 + \dots + \lambda_{n+1} \mathbf{A}_{n+1}$ means a **proper point** if $\sum_{k=1}^{n+1} \lambda_k \neq 0$ (namely the point with inhomogeneous barycentric coordinates $\rho \lambda_1, \dots, \rho \lambda_{n+1}$ with $\rho = (\sum_{k=1}^{n+1} \lambda_k)^{-1}$), or an **improper point** (determined by the (nonzero) vector $\lambda_1 \mathbf{A}_1 + \lambda_2 \mathbf{A}_2 + \dots + \lambda_{n+1} \mathbf{A}_{n+1}$) if $\sum_{k=1}^{n+1} \lambda_k = 0$. These coordinates can be viewed as in projective n -dimensional geometry (see Chapter 65). The set of improper points is, thus, characterized by the equation $\sum_{k=1}^{n+1} \lambda_k = 0$ in homogeneous barycentric coordinates, and is called the **improper hyperplane**.

The **circumcenter** of a simplex is the point that is equidistant from the all vertices of the simplex. The set of all points at that common distance from the circumcenter is the **circumscribed hypersphere**.

The $(n+1) \times (n+1)$ matrix $M = [m_{ij}]$ with m_{ij} equal to the square of the distance between the i th and j th vertex of an n -simplex Σ is called the **Menger matrix** of Σ .

The **Gramian** of an n -simplex $\Sigma = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{n+1}\}$ is defined as the Gram matrix Q of $n+1$ vectors $\mathbf{n}_1, \mathbf{n}_2, \dots, \mathbf{n}_{n+1}$ determined as follows: The first n of them form the biorthogonal pair with the n vectors $\mathbf{u}_1 = \mathbf{A}_1 - \mathbf{A}_{n+1}, \mathbf{u}_2 = \mathbf{A}_2 - \mathbf{A}_{n+1}, \dots, \mathbf{u}_n = \mathbf{A}_n - \mathbf{A}_{n+1}$, the remaining $\mathbf{n}_{n+1}$ is defined by $\mathbf{n}_{n+1} = -\sum_{i=1}^n \mathbf{n}_i$.

Facts:

1. There are $\binom{n+1}{2}$ edges $\{\mathbf{A}_i, \mathbf{A}_j\}$, $i \neq j$, in an n -simplex.
2. A 2-simplex is a segment and its circumcenter is its midpoint. The circumcenter of an n -simplex is the intersection of the lines ℓ_i , where ℓ_i is the line normal to face ω_i and through the circumcenter of ω_i .
3. [Blu53] (Menger, Schoenberg, etc.) The numbers m_{ij} , $i, j = 1, \dots, n+1$, can serve as squares of the lengths of edges (i.e., of distances between vertices) of an n -simplex if and only if $m_{ii} = 0$ for all i , and

$$\sum_{i,j} m_{ij} x_i x_j < 0, \quad \text{whenever} \quad \sum_i x_i = 0.$$

4. (Consequence of Fact 3) The matrix

$$M_0 = \begin{bmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & M \end{bmatrix},$$

where $\mathbf{1}$ is the column vector of all ones and $M = [m_{ij}]$ is a Menger matrix, is elliptic, i.e., has one eigenvalue positive and the remaining negative.

5. The Menger matrix is the matrix of the quadratic form $\sum_{i,j} m_{ij} x_i x_j$. If $x_1, \dots, x_{n+1}$ are considered as homogeneous barycentric coordinates of the point $\mathbf{X} = [x_1, \dots, x_{n+1}]$, then the equation

$$\sum_{i,j} m_{ij} x_i x_j = 0$$

is the equation of the circumscribed hypersphere of the n -simplex Σ . (Thus, the condition in Fact 3 can be interpreted that for all improper points, the value of the quadratic form $\sum_{i,j} m_{ij} x_i x_j$ is strictly negative.)

6. The n -dimensional volume V of an n -simplex Σ satisfies

$$V^2 = \frac{(-1)^{n-1}}{2^n (n!)^2} \det M_0,$$

where M_0 is the matrix in Fact 4 using the Menger matrix M of Σ .

7. The volume V_s of the n -simplex with vertices in $\mathbf{O}, \mathbf{A}_1, \dots, \mathbf{A}_n$ is equal to $V_s = \frac{1}{n!} V_p$, where V_p is the volume of the parallelepiped defined in Fact 4 of section 66.2 by n vectors $\mathbf{A}_i - \mathbf{O}$. Thus, $V_s = \frac{1}{n!} \sqrt{\det G(\mathbf{u}_1, \dots, \mathbf{u}_n)}$, where $G(\dots)$ means the Gram matrix and $\mathbf{u}_k = \mathbf{A}_k - \mathbf{O}$.
8. Let $Q_0 = \begin{bmatrix} q_{00} & \mathbf{q}_0^T \\ \mathbf{q}_0 & Q \end{bmatrix}$, where the matrix Q is the Gramian of the n -simplex Σ , the numbers in the column vector $\mathbf{q}_0$ are (-2) -multiples of the inhomogeneous barycentric coordinates of the circumcenter of Σ , and $\frac{1}{2}\sqrt{q_{00}}$ is the radius of the circumsphere of Σ . Then $M_0 Q_0 = -2I$.
9. The vectors $\mathbf{n}_1, \mathbf{n}_2, \dots, \mathbf{n}_{n+1}$ from the definition of the Gramian are the vectors of outer normals of the simplex, in some sense normalized.
10. The Gramian of every n -simplex is an $(n+1) \times (n+1)$ positive semidefinite matrix of rank n with row sums equal to zero. Conversely, every such matrix determines uniquely (apart from the position in the space) an n -simplex in E^n whose Gramian this matrix is.
11. Let Σ_S be a face of an n -simplex Σ determined by the vertices with index set $S \subset \{1, \dots, n+1\}$. Let Q be the Gramian of Σ . Then the Gramian of Σ_S is the Schur complement $Q/Q(S)$, where $Q(S)$ denotes the principal submatrix of Q corresponding to indices in the complement of S .

Examples:

1. Let us find the Menger matrix and the Gramian in the case of the segment $\{\mathbf{A}_1, \mathbf{A}_2\}$ of length d (i.e., a 1-simplex). The Menger matrix is $\begin{bmatrix} 0 & d^2 \\ d^2 & 0 \end{bmatrix}$. If $\mathbf{u}_1 = \mathbf{A}_1 - \mathbf{A}_2$, then $\mathbf{n}_1 = \frac{1}{d^2} \mathbf{u}_1$ since

$\langle \mathbf{u}_1, \mathbf{n}_1 \rangle$ has to be 1. Thus, the Gramian is the Gram matrix $G(\mathbf{n}_1, -\mathbf{n}_1)$, i.e., $\begin{bmatrix} \frac{1}{d^2} & -\frac{1}{d^2} \\ -\frac{1}{d^2} & \frac{1}{d^2} \end{bmatrix}$. Indeed,

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & d^2 \\ 1 & d^2 & 0 \end{bmatrix} \begin{bmatrix} d^2 & -1 & -1 \\ -1 & \frac{1}{d^2} & -\frac{1}{d^2} \\ -1 & -\frac{1}{d^2} & \frac{1}{d^2} \end{bmatrix} = -2I_3, \text{ as asserted in Fact 8.}$$

2. Consider the simplex $\Sigma = \{\mathbf{A}_1 = [0, 0, 1]^T, \mathbf{A}_2 = [1, 0, 1]^T, \mathbf{A}_3 = [1, 2, 1]^T\}$ in E^2 . We show how Fact 6 and Fact 7 can be used to find the volume V of this simplex.

To use Fact 6, compute the squares of the distances between the vertices to obtain the Menger

matrix: $M = \begin{bmatrix} 0 & 1 & 5 \\ 1 & 0 & 4 \\ 5 & 4 & 0 \end{bmatrix}$, so $M_0 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 5 \\ 1 & 1 & 0 & 4 \\ 1 & 5 & 4 & 0 \end{bmatrix}$. Since $\det M_0 = -16$, $V = 1$ by Fact 6.

To use Fact 7, compute the vectors $\mathbf{u}_1 = [1, 0, 0]^T$ and $\mathbf{u}_2 = [1, 2, 0]^T$, so $G(\mathbf{u}_1, \mathbf{u}_2) = \begin{bmatrix} 1 & 1 \\ 1 & 5 \end{bmatrix}$. By

Fact 7, $V = \frac{1}{2!} \sqrt{\det G(\mathbf{u}_1, \mathbf{u}_2)} = \frac{1}{2} \sqrt{4} = 1$.

In this example, the volume can be found directly by finding the area of the triangle.

3. Consider the simplex in Example 2. Let us compute the Gramian to illustrate Fact 8 for this simplex. Since $\mathbf{u}_1 = \mathbf{A}_1 - \mathbf{A}_3 = [-1, -2, 0]^T$, $\mathbf{u}_2 = \mathbf{A}_2 - \mathbf{A}_3 = [0, -2, 0]^T$, the vectors $\mathbf{n}_1, \mathbf{n}_2$ forming a biorthogonal pair with $\mathbf{u}_1, \mathbf{u}_2$ are (as in Example 2 of Section 66.2) $\mathbf{n}_1 = [-1, 0, 0]^T$, $\mathbf{n}_2 =$

$$[1, -\frac{1}{2}, 0]^T. \text{ Thus, } \mathbf{n}_3 = [0, \frac{1}{2}, 0]^T \text{ and } G(\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3) = \begin{bmatrix} 1 & -1 & 0 \\ -1 & \frac{5}{4} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}.$$

The line ℓ_1 is the set of points of the form $[x, 1, 1]^T$, and line ℓ_3 is the set of points of the form $[\frac{1}{2}, y, 1]^T$, so the circumcenter of the simplex is $\mathbf{c} = [\frac{1}{2}, 1, 1]^T$. The (inhomogeneous) barycentric coordinates b_1, b_2, b_3 of $\mathbf{c}$ can be found by solving $b_1 \mathbf{A}_1 + b_2 \mathbf{A}_2 + b_3 \mathbf{A}_3 = \mathbf{c}$, $b_1 + b_2 + b_3 = 1$

to obtain $b_1 = \frac{1}{2}, b_2 = 0, b_3 = \frac{1}{2}$. The radius of the circumsphere is $\frac{\sqrt{5}}{2}$, so $q_{00} = 5$. Thus,

$$Q_0 = \begin{bmatrix} 5 & -1 & 0 & -1 \\ -1 & 1 & -1 & 0 \\ 0 & -1 & \frac{5}{4} & -\frac{1}{4} \\ -1 & 0 & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}, \text{ which is indeed equal to the matrix } -2M_0^{-1}.$$

4. We can use either Example 1 or Fact 11 and Example 3 to find the Gramian of Σ_S for $\{2, 3\}$ (again with Σ defined in Example 2). By Example 1, the Gramian is $\begin{bmatrix} \frac{1}{d^2} & -\frac{1}{d^2} \\ -\frac{1}{d^2} & \frac{1}{d^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$.

66.4 Special Simplexes

Definitions:

Let us color an edge $\{A_i, A_j\}$ of an n -simplex with vertices $A_1, \dots, A_{n+1}$, $n \geq 2$, in: **red**, if the opposite interior angle ϕ_{ij} is acute; **blue**, if the opposite interior angle ϕ_{ij} is obtuse; it will stay **uncolored**, if the opposite interior angle ϕ_{ij} is right.

The edge $\{A_i, A_k\}$ is called **colored** if it is colored red or blue. The assignment of red and blue colors is a **coloring** of the simplex.

The **graph** of the n -simplex is the graph with vertices $1, 2, \dots, n+1$ and edges $\{i, k\}$, $i \neq k$, for which $\{A_i, A_k\}$ is colored. The **colored graph** of the simplex is the graph of the simplex colored in red and blue in the same way as the corresponding edges of the simplex.

An n -simplex is called **hypercute** if each of its dihedral interior angles ϕ_{ij} is either acute or right.

A hyperacute n -simplex is called **totally hyperacute** if its circumcenter is either an interior point of the simplex, or an interior point of one of its faces.

Let C denote the circumcenter of the n -simplex with vertices $A_1, \dots, A_{n+1}$. We extend the coloring of the simplex as follows: We assign the segment $\overline{CA_k}$ the **red** color if the point C is in the same open halfspace determined by the face ω_k as the vertex A_k , and the **blue** color, if C is in the opposite open halfspace. We do not assign to $\overline{CA_k}$ any color if C belongs to ω_k . This is the **extended coloring** of the simplex.

In the same way, we speak about the **extended graph** and **extended colored graph** of the simplex, adding to $n+1$ vertices another vertex $n+2$ which corresponds to the circumcenter.

A **right simplex** (cf. [Fie57]) is an n -simplex which has exactly n acute interior angles and all the remaining $\binom{n}{2}$ interior angles right.

In a right simplex, the edges opposite the acute angles are called **legs**. The subgraph of legs and their endpoints is called **tree of legs** (see Fact 5).

A right simplex whose tree of legs is a path is called a **Schlaefli simplex**.

The face of a right simplex spanned by all pendent vertices (i.e., vertices of degree one) of the tree of legs is called the **hypotenuse** of the simplex.

A **net** on a simplex is a subset of the set of edges (not necessarily connected) such that every vertex of the simplex belongs to some edge in the net. A net is **metric** if for each edge in the net its length is given.

A **box** in a Euclidean n -space is a parallelepiped all of whose edges at some vertex (and then at all vertices) are mutually perpendicular.

Facts (cf. [Fie57]):

1. A simplex is hyperacute if and only if it has no blue edge in its coloring.
2. The set of red edges connects all the vertices of the simplex.
3. If we color the edges of an n -simplex by the two colors red and blue in such a way that the red edges connect all vertices, then there exists such deformation of the simplex that opposite red edges there are acute, opposite blue edges obtuse, and opposite uncolored edges right interior angles.

4. Every n -simplex has at least n acute interior angles.
5. There exist right n -simplexes. The red edges span a tree containing all the vertices of the simplex.
6. The legs of a right simplex are mutually perpendicular.
7. The tree of legs of a right n -simplex can be completed to a $(n$ -dimensional) box; its center of symmetry is thus the circumcenter of the simplex. Conversely, every right n -simplex can be obtained by choosing among the edges of some box a subset of n edges with the property that any two are perpendicular and together form a connected set. These are then the legs of the simplex.
8. Let $G_T = (N, E, W)$ be a tree with the numbered vertex set $N = \{1, 2, \dots, n+1\}$ and edge set E ; let every edge $\{i, j\} \in E$ be assigned a positive number w_{ij} . Construct the matrices $Q_0 = [q_{rs}]$, $r, s = 0, 1, \dots, n+1$, and $M = [m_{ij}]$, $i, j = 1, \dots, n+1$, as follows:

$$q_{00} = \sum_{i,j \in E} \frac{1}{w_{ij}}, \quad q_{0i} = q_{i0} = d_i - 2, \quad i \in N,$$

where d_i is the degree of the vertex i in G_T ,

$$q_{ij} = -w_{ij} \text{ if } \{i, j\} \in E, \quad q_{ii} = \sum_{j, \{i, j\} \in E} w_{ij}, \quad q_{ij} = 0 \text{ otherwise;}$$

$$m_{ii} = 0, \quad i \in N, \quad m_{ij} = w_{ik_1}^{-1} + w_{k_1 k_2}^{-1} + \dots + w_{k_r j}^{-1},$$

where $i, j \in N$, $i \neq j$, and $i, k_1, k_2, \dots, k_r, j$ is the (unique) path in G_T from i to j .

Then the matrices Q_0 and $M_0 = \begin{bmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & M \end{bmatrix}$ satisfy

$$M_0 Q_0 = -2I_{n+2}$$

and they are matrices corresponding to a right n -simplex whose tree of legs is G_T .

9. The inhomogeneous barycentric coordinates of the circumcenter of a right n -simplex are $q_{0i} = 1 - \frac{1}{2}d_i$, where d_i is the degree of the vertex i in the tree of legs.
10. The hypotenuse of a Schläefli simplex is the longest edge; the midpoint of the hypotenuse is the circumcenter of the simplex.
11. Every face of a Schläefli simplex is also a Schläefli simplex.
12. The Schläefli simplex is the only simplex all 2-dimensional faces of which are right triangles.
13. An n -simplex Σ is a Schläefli simplex if and only if there exist real distinct numbers $c_1, \dots, c_{n+1}$ such that the Menger matrix $M = [m_{ij}]$ of Σ has the form $m_{ij} = |c_i - c_j|$. If this holds for $c_1 > c_2 > \dots > c_{n+1}$, then, in the usual notation, the edges $\{\mathbf{A}_k, \mathbf{A}_{k+1}\}$, $k = 1, \dots, n$, are the legs of Σ .
14. Let Σ be a hyperacute n -simplex. Then the Gramian Q of Σ is a singular M -matrix of rank n , with annihilating vector $\mathbf{1}$.
15. Every face of a hyperacute simplex Σ is also a hyperacute simplex. The distribution of acute and right angles in the face is completely determined by the distribution of acute and right angles in the simplex as follows: If the face is determined by vertices with indices in S , the edge $\{\mathbf{A}_i, \mathbf{A}_j\}$ in the face will be red if and only if one can proceed from the vertex $\mathbf{A}_i$ to the vertex $\mathbf{A}_j$ along a path in the set of red edges of Σ which does not contain any vertex with index in S (except $\mathbf{A}_i$ and $\mathbf{A}_j$).
16. ([Fie61]). In the extended colored graph of an n -simplex, $n \geq 2$, the red part has vertex connectivity at least two.
17. ([Fie61]). In the definition of the extended graph and extended coloring, the vertex $n+2$ has no privileged position in the following sense: Let G be the extended colored graph of some n -simplex Σ_1 , with vertex $n+2$ corresponding to the circumcenter of Σ_1 . If $k \in \{1, 2, \dots, n+1\}$, then there exists another n -simplex Σ_2 whose extended colored graph is G and such that k is the vertex corresponding to the circumcenter of Σ_2 .

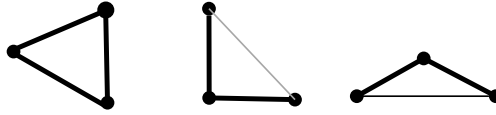


FIGURE 66.1 Coloring of triangles.

18. Every face (of dimension at least two) of a totally hyperacute simplex is also a totally hyperacute simplex. The extended coloring of the face is by the extended coloring of the simplex uniquely determined; it is obtained in the same way as in the usual coloring of a hyperacute simplex in Fact 15.
19. Let Σ be a totally hyperacute n -simplex. Then the extended graph of Σ is either a cycle, and then Σ is a Schläefli simplex, or it has vertex connectivity at least three.
20. ([Fie61]) Let a metric net N on a simplex be given. There exists a simplex of maximum volume with this net if and only if the net N is connected, i.e., if it is possible to pass from any vertex of the net to any other vertex using edges in the net only. In addition, every simplex with this maximum volume has the property that every interior angle opposite an unspecified edge of the simplex (i.e., not belonging to N) is right.

Examples:

1. Figure 66.1 shows examples of three colored triangles. In these diagrams, the heaviest line represents red, the ordinary line represents blue, and the light gray line represents white.
2. Figure 66.2 shows examples of the two types of right tetrahedra. The dashed lines indicate the box described in Fact 7. The tetrahedron on the right is a Schläefli simplex.
3. Figure 66.3 shows examples of extended graphs of triangles in Figure 66.1.
4. We apply the result in Fact 20 to so-called **cyclic simplexes** (cf. [Fie61]), which are maximum volume simplexes in the case that the metric net is a cycle. We call a simplex **regularly cyclic** if all edges in this net have the same length.

The Gramian of the regularly cyclic n -simplex is a multiple of the matrix

$$\begin{bmatrix} 2 & -1 & 0 & \cdots & 0 & -1 \\ -1 & 2 & -1 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 2 & -1 \\ -1 & 0 & 0 & \cdots & -1 & 2 \end{bmatrix}$$

with $n + 1$ rows and columns, if the net is formed by the edges $\{\mathbf{A}_k, \mathbf{A}_{k+1}\}$, $k = 1, \dots, n$, and $\{\mathbf{A}_{n+1}, \mathbf{A}_1\}$.

The corresponding Menger matrix is then proportional to the matrix with entries $m_{ik} = |i - k|(n + 1 - |i - k|)$. It is possible to show that any two of the edges in the cycle span the same angle.

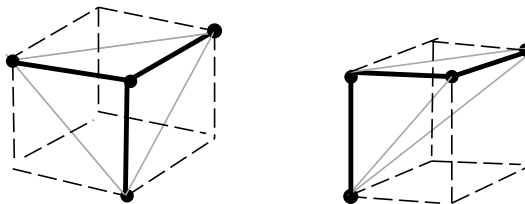


FIGURE 66.2 The two types of right tetrahedra.

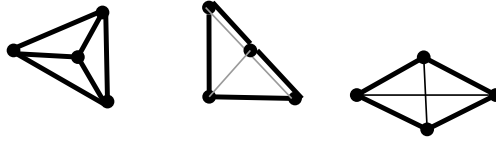


FIGURE 66.3 Extended graphs of triangles.

It is immediate that the regularly cyclic 2-simplex is the equilateral triangle. The regularly cyclic 3-simplex is the tetrahedron, which is obtained from a square by parallelly lifting one diagonal in the perpendicular direction to the plane so that the distance of the two new diagonals equals half of the length of each diagonal.

Thus, the volume of every n -simplex with vertices $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{n+1}$ in a Euclidean n -space for which all the edges $\{\mathbf{A}_1, \mathbf{A}_2\}, \{\mathbf{A}_2, \mathbf{A}_3\}, \dots, \{\mathbf{A}_n, \mathbf{A}_{n+1}\}, \{\mathbf{A}_{n+1}, \mathbf{A}_1\}$ have length one, does not exceed $\frac{1}{n!} \sqrt{\frac{(n+1)^{n-1}}{n^n}}$.

66.5 An Application to Resistive Electrical Networks

We conclude with applications of matrices and graphs in another, maybe surprising, field.

Definitions:

A **resistive electrical network** is a network consisting of a finite number of nodes, say $1, 2, \dots, n$, some pairs of which are directly connected by a conductor of some resistance. The assumptions are that the whole network is connected, and that there are no other electrical elements than resistors. As usual, **conductivity** of the conductor between two nodes is the reciprocal of the resistance of that conductor. If there is no direct connection between a pair of nodes, we set the conductivity of that “conductor” as zero.

A resistive electrical network of which just some of the nodes, **outlets**, are accessible, is called a **black-box**. The matrix of mutual resistances between the outlets of a black-box is called a **black-box matrix**.

The **problem** is: Characterize the set of all possible black-box $n \times n$ matrices.

Facts ([Moo68], [Fie78]):

1. Resistances of conductors in series add and conductivities of conductors in parallel add. If conductors having resistances R_1, R_2 are placed in series (see left illustration in Figure 66.4) the resistance R between the nodes is $R_1 + R_2$. If conductors having resistances R_1, R_2 are placed in parallel (see right illustration in Figure 66.4) the resistance R between the nodes satisfies $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$.
2. The 3×3 black-box matrices are all matrices

$$\begin{bmatrix} 0 & r_{12} & r_{13} \\ r_{12} & 0 & r_{23} \\ r_{13} & r_{23} & 0 \end{bmatrix}$$

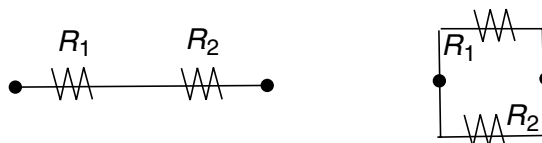


FIGURE 66.4 Resistors in series and parallel.

in which the numbers r_{12} , r_{13} , and r_{23} are nonnegative, at least two different from zero, and fulfill the nonstrict triangle inequality.

3. The $n \times n$ black-box matrices are exactly Menger matrices of hyperacute $(n - 1)$ -simplexes.
4. If a black-box matrix is given, then the corresponding network can be realized as such for which the conductivities between pairs of outlets are equal to the negatively taken corresponding entries of the Gramian of the simplex whose Menger matrix the given matrix is.
5. ([Fie78]) Let the given black-box B with n outlets correspond to an $(n - 1)$ -simplex Σ . Let S be a nonvoid proper subset of the set of outlets of B . Join all outlets in S by shortcuts. The resulting device can be considered as a new black-box B_S , which has just one outlet instead those outlets in S and, of course, all the remaining outlets. We construct the simplex corresponding to B_S and its black-box matrix as follows.

If L is the linear space in the corresponding E^{n-1} determined by the vertices corresponding to S , project orthogonally all the remaining vertices as well as L itself on (some) orthogonal complement $L^\perp$ to L , thus obtaining a new simplex. The Menger matrix M_S of this simplex will be the black-box matrix of B_S . An algebraic construction is to pass from the Menger matrix of Σ , to the Gramian Q of Σ using Fact 8 of section 66.3; in this Gramian Q , add together all the rows corresponding to S into a single row and all the columns corresponding to S into a single column. The resulting symmetric matrix $\hat{Q}$ is again a singular M -matrix with row-sums zero. The simplex whose Gramian is $\hat{Q}$ is then that whose Menger matrix is B_S .

6. Let S_1 and S_2 be two disjoint nonempty subsets of the set of outlets of a black-box, and let Σ be the simplex in the geometric model. Join all outlets in S_1 with a source of potential zero, and all outlets in S_2 with the source of potential one. What will be the distribution of potentials in the remaining outlets using the geometric interpretation?

The answer is: Let L_1 , L_2 , respectively, be linear spaces determined by vertices of Σ corresponding to S_1 , S_2 , respectively. There exists a unique pair of parallel hyperplanes H_1 and H_2 , such that H_1 contains L_1 and H_2 contains L_2 , and the distance between H_1 and H_2 is maximal among all such pairs of parallel hyperplanes. Then the square of the distance of the hyperplanes H_1 and H_2 measures the resistance between S_1 and S_2 (similarly like the square of the distance between two vertices of Σ measures the resistance between the corresponding outlets), and if V_0 is a vertex corresponding to an outlet S_0 , then the potential in S_0 is obtained by linear interpolation corresponding to the position of the hyperplane H_0 containing V_0 and parallel to H_1 with respect to the hyperplanes H_1 and H_2 . Let us remark that the whole simplex Σ is in the layer between H_1 and H_2 , thanks to the property of hyperacuteness of Σ .

Examples:

1. Suppose the three nodes 1, 2, and 3 are connected as follows: The nodes 1 and 2 are connected by a conductor of resistance 18 ohms, the nodes 1 and 3 by conductor of resistance 12 ohms, and the nodes 2 and 3 by conductor of resistance 6 ohms. We compute the resistances for the black-box with the three outlets 1, 2, and 3. For r_{12} , the total resistance between outlets 1 and 2, note that there are two parallel paths between 1 and 2: the direct path with resistance 18 ohms, and the path through node 3 with resistance 6 ohms + 12 ohms = 18 ohms. Thus $\frac{1}{r_{12}} = \frac{1}{18} + \frac{1}{18} = \frac{1}{9}$, so $r_{12} = 9$ ohms. Similar computations show that the resistance between outlets 1 and 3 is 8 ohms and the

resistance between outlets 2 and 3 is 5 ohms. The black-box matrix is
$$\begin{bmatrix} 0 & 9 & 8 \\ 9 & 0 & 5 \\ 8 & 5 & 0 \end{bmatrix}.$$

2. Let us remark that the properties of the outlets of a black-box do not depend on the way the conductors and resistors in the box are set. Here is another way the black-box-matrix in Example 1 could be obtained: There are no direct connections between nodes 1, 2, and 3, but there is a fourth node 4, which is connected with node 1 by a conductor of resistance 6 ohms, with node 2 by a conductor of resistance 3 ohms, and with node 3 by a conductor of resistance 2 ohms. Since

resistors in series add, this produces the same black-box matrix. If we then connect the outlets 1 and 2 by a short-circuit (conductor of no resistance), in both cases the resulting resistance between the new outlet $\{1, 2\}$ and outlet 3 will be the same, equal to 4 ohms.

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Applications to Algebra

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67

Matrix Groups

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The topics of this chapter and the next (on group representations) are closely related. Here we consider some particular groups that arise most naturally as matrix groups or quotients of them, and special properties of matrix groups that are not shared by arbitrary groups. In representation theory, we consider what we learn about a group by considering all its homomorphisms to matrix groups. In this chapter we discuss properties of specific matrix groups, especially the general linear group (consisting of all invertible matrices of given size over a given field) and the related “classical groups.” Most group theoretic terminology is standard and can be found in any textbook or in the Preliminaries in the Front Matter of the book.

67.1 Introduction

Definitions:

The **general linear group** $GL(n, F)$ is the group consisting of all invertible $n \times n$ matrices over the field F .

A **matrix group** is a subgroup of $GL(n, F)$ for some natural number n and field F .

If V is a vector space of dimension n over F , the **group of invertible linear operators** on V is denoted by $GL(V)$.

A **linear group of degree n** is a subgroup of $GL(V)$ (where $\dim V = n$). (A subgroup of $GL(n, F)$ is a linear group of degree n , since an $n \times n$ matrix can be viewed as a linear operator acting on F^n by matrix multiplication.)

A linear group $G \leq GL(V)$ is said to be **reducible** if there is a G -invariant subspace U of V other than $\{0\}$ and V , and is **irreducible** otherwise. If $G \leq GL(V)$ is irreducible, then V is called **G -irreducible**.

A linear group $G \leq GL(V)$ is said to be **decomposable** if V is the direct sum of two nonzero G -invariant subspaces, and is **indecomposable** otherwise.

If V can be expressed as the direct sum of G -irreducible subspaces, then G is **completely reducible**. (An irreducible group is completely reducible.)

If the matrix group $G \leq GL(n, F)$ is irreducible regarded as a subgroup of $GL(n, K)$ for any algebraic extension K of F , we say that G is **absolutely irreducible**.

A linear group of degree n is **unipotent** if all its elements have n eigenvalues equal to 1.

Let X and Y be group-theoretic properties.

A group G is **locally X** if every finite subset of G is contained in a subgroup with property X .

Facts:

For these facts and general background reading see [Dix71], [Sup76], and [Weh73].

1. If V is a vector space of dimension n over F , then $\text{GL}(V)$ is isomorphic to $\text{GL}(n, F)$.
2. Every finite group is isomorphic to a matrix group.
3. Basic facts from linear algebra about similarity of matrices can be interpreted as statements about conjugacy classes in $\text{GL}(n, F)$. For example:
 - Two nonsingular matrices are conjugate in $\text{GL}(n, F)$ if and only if they have the same invariant factors.
 - If F is algebraically closed, then two nonsingular matrices are conjugate in $\text{GL}(n, F)$ if and only if they have the same Jordan canonical form.
 - Two real symmetric matrices are conjugate in $\text{GL}(n, \mathbb{R})$ if and only if they have the same rank and signature.

(See Chapter 6 for more information on the Jordan canonical form and invariant factors and Chapter 12 for more information on signature.)
4. A matrix group G of degree n is reducible if and only if there exists a nonsingular matrix $M \in F^{n \times n}$ and k with $1 \leq k \leq n - 1$ such that for all $A \in G$, $M^{-1}AM$ is of the form $\begin{bmatrix} B_{11} & B_{12} \\ \mathbf{0} & B_{22} \end{bmatrix}$, where $B_{11} \in F^{k \times k}$, $B_{22} \in F^{(n-k) \times (n-k)}$.
5. (See Chapter 68) The image of a representation of a group is a linear group. The image of a matrix representation of a group is a matrix group. We apply descriptions of the linear group to the representation: If $\rho : G \rightarrow \text{GL}(V)$ is a representation, and $\rho(G)$ is irreducible, indecomposable, absolutely irreducible, etc., then we say that the representation ρ is irreducible, etc.
6. If every finitely generated subgroup of a group G is isomorphic to a linear group of degree n over a field F (of arbitrary characteristic), then G is isomorphic to a linear group of degree n .
7. Any free group is linear of degree 2 in every characteristic. More generally, a free product of linear groups is linear.
8. (Maschke's Theorem) Let G be a finite linear group over F , and suppose that the characteristic of F is either zero or coprime to $|G|$. If G is reducible, then it is decomposable.
9. A locally finite linear group in characteristic zero is completely reducible.
10. (Clifford's Theorem) Let G be an irreducible linear group on a vector space V of dimension n , and let N be a normal subgroup of G . Then V is a direct sum of minimal N -spaces $W_1, \dots, W_d$ permuted transitively by G . In particular, d divides n , the group N is completely reducible, and the linear groups induced on W_i by N are all isomorphic.
11. A normal (or even a subnormal) subgroup of a completely reducible linear group is completely reducible.
12. A unipotent matrix group is conjugate (in the general linear group) to the group of upper unit triangular matrices.
13. A linear group G on V has a unipotent normal subgroup U such that G/U is isomorphic to a completely reducible linear group on V ; the subgroup U is a nilpotent group of class at most $n - 1$, where $n = \dim(V)$.
14. (Mal'cev) If every finitely generated subgroup of the linear group G is completely reducible, then G is completely reducible.
15. Let G be a linear group on F^n , where F is algebraically closed. Then G is irreducible if and only if the elements of G span the space $F^{n \times n}$ of all $n \times n$ matrices over F .

Examples:

1. The matrix group

$$G = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{4}{3} & -\frac{1}{3} \end{bmatrix}, \begin{bmatrix} -\frac{1}{3} & -\frac{2}{3} \\ -\frac{4}{3} & \frac{1}{3} \end{bmatrix} \right\}$$

is decomposable and completely reducible: the subspaces of $\mathbb{R}^2$ spanned by the vectors $[1, 1]^T$ and $[-1, 2]^T$ are G -invariant. That is, with $M = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$, for any $A \in G$, $M^{-1}AM$ is a diagonal matrix.

2. The matrix group

$$\left\{ \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} : a \in \mathbb{R} \right\}$$

is reducible, but neither decomposable nor completely reducible.

3. The matrix group

$$\left\{ \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} : x \in \mathbb{R} \right\}$$

of real rotations is irreducible over $\mathbb{R}$ but not over $\mathbb{C}$: The subspace spanned by $[1, i]^T$ is invariant. The matrices in this group span the 2-dimensional subspace of $\mathbb{R}^{2 \times 2}$ consisting of matrices $A = [a_{ij}]$ satisfying the equations $a_{11} = a_{22}$ and $a_{12} + a_{21} = 0$.

4. A group is locally finite if and only if every finitely generated subgroup is finite.

67.2 The General and Special Linear Groups

Definitions:

The **special linear group** $\text{SL}(n, F)$ is the subgroup of $\text{GL}(n, F)$ consisting of matrices of determinant 1. If V is a vector space of dimension n over F , the group of invertible linear operators on V having determinant 1 is denoted by $\text{SL}(V)$.

The **special linear group** $\text{SL}(n, F)$ is the subgroup of $\text{GL}(n, F)$ consisting of matrices of determinant 1.

The **projective general linear group** and **projective special linear group** $\text{PGL}(n, F)$ and $\text{PSL}(n, F)$ are the quotients of $\text{GL}(n, F)$ and $\text{SL}(n, F)$ by their normal subgroups Z and $Z \cap \text{SL}(n, F)$, respectively, where Z is the group of nonzero scalar matrices.

Notation: If $F = \text{GF}(q)$ is the finite field of order q , then $\text{GL}(n, F)$ is denoted $\text{GL}(n, q)$, $\text{SL}(n, F)$ is denoted $\text{SL}(n, q)$, etc.

A **transvection** is a linear operator T on V with all eigenvalues equal to 1 and satisfying $\text{rank}(T - I) = 1$.

Facts:

For these facts and general background reading see [HO89], [Tay92], or [Kra02].

1. The special linear group $\text{SL}(n, F)$ is a normal subgroup of the general linear group $\text{GL}(n, F)$.
2. The order of $\text{GL}(n, q)$ is equal to the number of ordered bases of $\text{GF}(q)^n$, namely

$$|\text{GL}(n, q)| = \prod_{i=0}^{n-1} (q^n - q^i) = q^{n(n-1)/2} \prod_{i=0}^{n-1} (q^{n-i} - 1).$$

3. A transvection has determinant 1, and so lies in $\text{SL}(V)$.
4. A transvection on F^n has the form $I + \mathbf{v}\mathbf{w}^T$ for some $\mathbf{v}, \mathbf{w} \in F^n$ and $\mathbf{w}^T \mathbf{v} = 0$, and anything in this form is a transvection.
5. A transvection T on V has the form $T : x \mapsto x + f(x)\mathbf{v}$, where $\mathbf{v} \in V$, $f \in V^*$, and $f(\mathbf{v}) = 0$, and anything in this form is a transvection.
6. The group $\text{SL}(n, F)$ is generated by transvections, for any $n \geq 2$ and any field F .

7. The group $\text{PSL}(n, F)$ is simple for all $n \geq 2$ and all fields F , except for the two cases $\text{PSL}(2, 2)$ and $\text{PSL}(2, 3)$. The groups $\text{PSL}(2, 2)$ and $\text{PSL}(2, 3)$ are isomorphic to the symmetric group on 3 letters and the alternating group on 4 letters, respectively. The groups $\text{PSL}(2, 4)$ and $\text{PSL}(2, 5)$ are both isomorphic to the alternating group on 5 letters.

Examples:

1. $\text{GL}(2, 2) = \text{SL}(2, 2) = \text{PSL}(2, 2) =$

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}.$$

2. $\text{SL}(2, 3) =$

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}, \right. \\ \left. \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \right\}.$$

- 3.

$$T = \begin{bmatrix} -13 & -10 & -2 \\ 14 & 11 & 2 \\ 28 & 20 & 5 \end{bmatrix}$$

is a transvection. $T = I + \mathbf{v}\mathbf{w}^T$ with $\mathbf{v} = [-2, 2, 4]^T$, and $\mathbf{w} = [7, 5, 1]^T$.

67.3 The BN Structure of the General Linear Group

Definitions:

A **BN-pair** (or **Tits system**) is an ordered quadruple (G, B, N, S) where

- G is a group generated by subgroups B and N .
- $T := B \cap N$ is normal in N .
- S is a subset of $W := N/T$ and S generates W .
- The elements of S are all of order 2.
- If $\rho, \sigma \in N$ and $\rho T \in S$, then $\rho B \sigma \subseteq B \sigma B \cup B \rho \sigma B$.
- If $\rho T \in S$, then $\rho B \rho \neq B$.

If (G, B, N, S) is a BN-pair, the subgroups B and $W = N/T$ are known as the **Borel subgroup** and **Weyl group** of G .

A **parabolic subgroup** of G (relative to a given BN-pair) is a subgroup of the form $P_I = \langle B, s_i : i \in I \rangle$ for some subset I of $\{1, \dots, |S|\}$.

Facts:

For these facts and general background reading see [HO89], [Tay92], or [Kra02].

1. The general linear group $\text{GL}(n, F)$ with $n \geq 2$ has the following Tits system:
 - B is the group of upper-triangular matrices in G .
 - U the group of unit upper-triangular matrices (with diagonal entries 1).

- T the group of diagonal matrices.
 - N is the group of matrices having a unique nonzero element in each row or column.
 - $S = \{s_i : i = 1, \dots, n-1\}$, where $s_i = P_i T$ and P_i is the reflection which interchanges the i th and $(i+1)$ st standard basis vectors (P_i is obtained from the identity matrix by interchanging rows i and $i+1$).
 - N is the normalizer of T in $\mathrm{GL}(n, F)$.
 - $B = UT$.
 - $B \cap N = T$.
 - N/T is isomorphic to the symmetric group S_n .
2. If G has a BN-pair, any subgroup of G containing B is a parabolic subgroup.
 3. In $\mathrm{GL}(n, F)$, there are 2^{n-1} parabolic subgroups for the BN-pair in Fact 1, hence, there are 2^{n-1} subgroups of $\mathrm{GL}(n, F)$ containing the subgroup B of upper-triangular matrices.
 4. More generally, with respect to any basis of V there is a BN-structure. The terms *Borel subgroup* and *parabolic subgroup* are used to refer to the subgroups defined with respect to an arbitrary basis. All the Borel subgroups of $\mathrm{GL}(V)$ are conjugate. The maximal parabolic subgroups are precisely the maximal reducible subgroups.

Examples:

1. The maximal parabolic subgroups of $\mathrm{GL}(n, F)$ with the BN-pair in Fact 1 are those for which $I = \{1, \dots, n-1\} \setminus \{k\}$ for some k ; it is easy to see that in this case P_I is the stabilizer of the subspace spanned by the first k basis vectors. This subgroup consists of all matrices with block form as in 67.1, Fact 4.

67.4 Classical Groups

The classical groups form several important families of linear groups. We give a brief description here, and refer to the books [HO89], [Tay92], or the article [Kra02] for more details. For information on bilinear, sesquilinear, and quadratic forms, see Chapter 12.

Definitions:

A φ -sesquilinear form B is φ -**Hermitian** if $B(\mathbf{v}, \mathbf{w}) = \varphi(B(\mathbf{w}, \mathbf{v}))$ for all $\mathbf{v}, \mathbf{w} \in V$. In the case where $F = \mathbb{C}$ and φ is conjugation, a φ -Hermitian form is called a **Hermitian** form.

A **formed space** is a finite dimensional vector space carrying a nondegenerate φ -Hermitian, symmetric, or alternating form B .

A **classical group** over a formed space V is the subgroup of $\mathrm{GL}(V)$ consisting of the linear operators that preserve the form. We distinguish three types of classical groups:

1. **Orthogonal group:** Preserving a nondegenerate symmetric bilinear form B .
2. **Symplectic group:** Preserving a nondegenerate alternating bilinear form B .
3. **Unitary group:** Preserving a nondegenerate σ -Hermitian form B , with $\sigma \neq 1$.

We denote a classical subgroup of $\mathrm{GL}(V)$ by $\mathrm{O}(V)$, $\mathrm{Sp}(V)$, or $\mathrm{U}(V)$ depending on type. If necessary, we add extra notation to specify which particular form is being used. If $V = F^n$, we also write $\mathrm{O}(n, F)$, $\mathrm{Sp}(n, F)$, or $\mathrm{U}(n, F)$.

The **Witt index** of a formed space V is the dimension of the largest subspace on which the form is identically zero. The Witt index of the corresponding classical group is the Witt index of the formed space.

An **isometry** between subspaces of a formed space is a linear transformation preserving the value of the form.

A representation of a group G over the complex numbers is said to be **unitary** if its image is contained in the unitary group.

Facts:

For these facts and for general background reading see [HO89], [Tay92], or [Kra02].

1. The only automorphism of $\mathbb{R}$ is the identity, so any sesquilinear form on a real vector space is bilinear and any φ -Hermitian form is symmetric. A real formed space has a symmetric or alternating bilinear form as its form. The classical subgroups of a real formed space are orthogonal or symplectic.
2. The only automorphisms of $\mathbb{C}$ that preserve the reals are the identity and complex conjugation. Any φ -Hermitian form such that φ preserves the reals is a Hermitian form.
3. Classification of classical groups up to conjugacy in $\mathrm{GL}(n, F)$ is equivalent to classification of forms of the appropriate type up to the natural action of the general linear group together with scalar multiplication. Often this is a very difficult problem; the next fact gives a few cases where the classification is more straightforward.
4. (a) A nondegenerate alternating form on $V = F^n$ exists if and only if n is even, and all such forms are equivalent. So there is a unique conjugacy class of symplectic groups in $\mathrm{GL}(n, F)$ if n is even (with Witt index $n/2$), and none if n is odd.
 (b) Let $F = \mathrm{GF}(q)$. Then, up to conjugacy, $\mathrm{GL}(n, q)$ contains one conjugacy class of unitary subgroups (with Witt index $\lfloor n/2 \rfloor$), one class of orthogonal subgroups if n is odd (with Witt index $(n-1)/2$), and two classes if n is even (with Witt indices $n/2$ and $n/2-1$).
 (c) A nondegenerate symmetric bilinear form on $\mathbb{R}^n$ is determined up to the action of $\mathrm{GL}(n, \mathbb{R})$ by its signature. Its Witt index is $\min\{s, t\}$, where s and t are the numbers of positive and negative eigenvalues. So there are $\lfloor n/2 \rfloor + 1$ conjugacy classes of orthogonal subgroups of $\mathrm{GL}(n, \mathbb{R})$, with Witt indices $0, 1, \dots, \lfloor n/2 \rfloor$.
5. (Witt's Lemma) Suppose that U_1 and U_2 are subspaces of the formed space V , and $h : U_1 \rightarrow U_2$ is an isometry. Then there is an isometry g of V that extends h .
6. From Witt's Lemma it is possible to write down formulas for the orders of the classical groups over finite fields similar to the formula in Fact 2 general linear group.
7. The analogues of Facts 6 and 7 in section 67.2 hold for the classical groups with nonzero Witt index. However, the situation is more complicated. Any symplectic transformation has determinant 1, so $\mathrm{Sp}(2r, F) \leq \mathrm{SL}(2r, F)$. Moreover, $\mathrm{Sp}(2r, F)$ is generated by symplectic transvections (those preserving the alternating form) for $r \geq 2$, except for $\mathrm{Sp}(4, 2)$. Similarly, the **special unitary group** $\mathrm{SU}(n, F)$ (the intersection of $\mathrm{U}(n, F)$ with $\mathrm{SL}(n, F)$) with positive Witt index is generated by unitary transvections (those preserving the Hermitian form), except for $\mathrm{SU}(3, 2)$. Results for orthogonal groups are more difficult. See [HO89] for more information.
8. Like the general linear groups, the classical groups contain BN-pairs (configurations of subgroups satisfying conditions like those in the previous section). The difference is that the Weyl group $W = N/H$ is not the symmetric group, but one of the other types of Coxeter group (finite groups generated by reflections).
9. Although this treatment of classical groups has been as far as possible independent of fields, for most of mathematics, the classical groups over the real and complex numbers are the most important, and among these, the real orthogonal and complex unitary groups preserving positive definite forms most important; see [Wey39].
10. The theory can be extended to classical groups over rings. This has important connections with algebraic K-theory. The book [HO89] gives details.
11. Every representation of a finite group is equivalent to a unitary representation.

Examples:

1. The function B given by $B((x_1, y_1), (x_2, y_2)) = x_1 y_2 - x_2 y_1$ is an alternating bilinear form on F^2 . Any matrix with determinant 1 will preserve this form. So $\mathrm{Sp}(2, F) = \mathrm{SL}(2, F)$.
2. The symmetric group S_6 acts on F^6 , where F is the field with two elements. It preserves the 1-dimensional subspace U spanned by $(1, 1, 1, 1, 1, 1)$, as well as the 5-dimensional subspace consisting of vectors with coordinate sum zero. The usual dot product on F^6 is alternating when

restricted to W , and its radical is U , so it induces a symplectic form on W/U . Thus S_6 is a subgroup of the symplectic group $Sp(4, 2)$. Since both groups have order 720, we see that $Sp(4, 2) = S_6$.

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68

Group Representations

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Representation theory is the study of the various ways a given group can be mapped into a general linear group. This information has proven to be effective at providing insight into the structure of the given group as well as the objects on which the group acts. Most notable is the central contribution made by representation theory to the complete classification of finite simple groups [Gor94]. (See also Fact 3 of Section 68.1 and Fact 5 of Section 68.6.)

Representations of finite groups can be defined over an arbitrary field and such have been studied extensively. Here, however, we discuss only the most widely used, classical theory of representations over the field of complex numbers (many results of which fail to hold over other fields).

68.1 Basic Concepts

Throughout, G denotes a finite group, e denotes its identity element, and V denotes a finite dimensional complex vector space.

Definitions:

The **general linear group** of a vector space V is the group $GL(V)$ of linear isomorphisms of V onto itself with operation given by function composition.

A **(linear) representation** of the finite group G (over the complex field $\mathbb{C}$) is a homomorphism $\rho = \rho_V : G \rightarrow GL(V)$, where V is a finite dimensional vector space over $\mathbb{C}$.

The **degree** of a representation ρ_V is the dimension of the vector space V .

Two representations $\rho : G \rightarrow GL(V)$ and $\rho' : G \rightarrow GL(V')$ are **equivalent** (or **isomorphic**) if there exists a linear isomorphism $\tau : V \rightarrow V'$ such that $\tau \circ \rho(s) = \rho'(s) \circ \tau$ for all $s \in G$.

Given a representation ρ_V of G , a subspace W of V is **G -stable** (or **G -invariant**) if $\rho_V(s)(W) \subseteq W$ for all $s \in G$.

If ρ_V is a representation of G and W is a G -stable subspace of V , then the induced maps $\rho_W : G \rightarrow GL(W)$ and $\rho_{V/W} : G \rightarrow GL(V/W)$ are the corresponding **subrepresentation** and **quotient representation**, respectively.

A representation ρ_V of G with $V \neq \{0\}$ is **irreducible** if V and $\{0\}$ are the only G -stable subspaces of V ; otherwise, ρ_V is **reducible**.

The **kernel** of a representation ρ_V of G is the set of all $s \in G$ for which $\rho_V(s) = 1_V$.

A representation of G is **faithful** if its kernel consists of the identity element alone.

An **action** of G on a set X is a function $G \times X \rightarrow X$, $(s, x) \mapsto sx$, satisfying

- $(st)x = s(tx)$ for all $s, t \in G$ and $x \in X$,
- $ex = x$ for all $x \in X$.

A **$\mathbb{C}G$ -module** is a finite-dimensional vector space V over $\mathbb{C}$ together with an action $(s, v) \mapsto sv$ of G on V that is linear in the variable v , meaning

- $s(v + w) = sv + sw$ for all $s \in G$ and $v, w \in V$,
- $s(\alpha v) = \alpha(sv)$ for all $s \in G$, $v \in V$ and $\alpha \in \mathbb{C}$.

(See Fact 6 below.)

Facts:

The following facts can be found in [Isa94, pp. 4–10] or [Ser77, pp. 3–13, 47].

1. If $\rho = \rho_V$ is a representation of G , then
 - $\rho(e) = 1_V$,
 - $\rho(st) = \rho(s)\rho(t)$ for all $s, t \in G$,
 - $\rho(s^{-1}) = \rho(s)^{-1}$ for all $s \in G$.
2. A representation of G of degree one is a group homomorphism from G into the group $\mathbb{C}^\times$ of nonzero complex numbers under multiplication (identifying $\mathbb{C}^\times$ with $GL(\mathbb{C})$). Every representation of degree one is irreducible.
3. The group G is abelian if and only if every irreducible representation of G is of degree one.
4. *Maschke's Theorem:* If ρ_V is a representation of G and W is a G -stable subspace of V , then there exists a G -stable vector space complement of W in V .
5. *Schur's Lemma:* Let $\rho : G \rightarrow GL(V)$ and $\rho' : G \rightarrow GL(V')$ be two irreducible representations of G and let $f : V \rightarrow V'$ be a linear map satisfying $f \circ \rho(s) = \rho'(s) \circ f$ for all $s \in G$.
 - If ρ' is not equivalent to ρ , then f is the zero map.
 - If $V' = V$ and $\rho' = \rho$, then f is a scalar multiple of the identity map: $f = \alpha 1_V$ for some $\alpha \in \mathbb{C}$.
6. If $\rho = \rho_V$ is a representation of G , then V becomes a $\mathbb{C}G$ -module with action given by $sv = \rho(s)(v)$ ($s \in G$, $v \in V$). Conversely, if V is a $\mathbb{C}G$ -module, then $\rho_V(s)(v) = sv$ defines a representation $\rho_V : G \rightarrow GL(V)$ (called the **representation of G afforded by V**). The study of representations of the finite group G is the same as the study of $\mathbb{C}G$ -modules.
7. The vector space $\mathbb{C}G$ over $\mathbb{C}$ with basis G is a ring (the **group ring** of G over $\mathbb{C}$) with multiplication obtained by linearly extending the operation in G to arbitrary products. If V is a $\mathbb{C}G$ -module and the action of G on V is extended linearly to a map $\mathbb{C}G \times V \rightarrow V$, then V becomes a (left unitary) $\mathbb{C}G$ -module in the ring theoretic sense, that is, V satisfies the usual vector space axioms (see Section 1.1) with the scalar field replaced by the ring $\mathbb{C}G$.

Examples:

See also examples in the next section.

1. Let $n \in \mathbb{N}$ and let $\omega \in \mathbb{C}$ be an n th root of unity (meaning $\omega^n = 1$). Then the map $\rho : \mathbb{Z}_n \rightarrow \mathbb{C}^\times$ given by $\rho(m) = \omega^m$ is a representation of degree one of the group $\mathbb{Z}_n$ of integers modulo n . It is irreducible.
2. *Regular representation:* Let $V = \mathbb{C}G$ be the complex vector space with basis G . For each $s \in G$ there is a unique linear map $\rho(s) : V \rightarrow V$ satisfying $\rho(s)(t) = st$ for all $t \in G$. Then $\rho : G \rightarrow GL(V)$ is a representation of G called the **(left) regular representation**. If $|G| > 1$, then the regular representation is reducible (see Example 3 of Section 68.5).

3. *Permutation representation*: Let X be a finite set, let $(s, x) \mapsto sx$ be an action of G on X , and let V be the complex vector space with basis X . For each $s \in G$ there is a unique linear map $\rho(s) : V \rightarrow V$ satisfying $\rho(s)(x) = sx$ for all $x \in X$. Then $\rho : G \rightarrow GL(V)$ is a representation of G called a **permutation representation**. The regular representation of G (Example 2) is the permutation representation corresponding to the action of G on itself given by left multiplication.
4. The representation of G of degree 1 given by $\rho(s) = 1 \in \mathbb{C}^\times$ for all $s \in G$ is the **trivial representation**.
5. *Direct sum*: If V and W are $\mathbb{C}G$ -modules, then the $\mathbb{C}$ -vector space direct sum $V \oplus W$ is a $\mathbb{C}G$ -module with action given by $s(v, w) = (sv, sw)$ ($s \in G, v \in V, w \in W$).
6. *Tensor product*: If V_1 is a $\mathbb{C}G_1$ -module and V_2 is a $\mathbb{C}G_2$ -module, then the $\mathbb{C}$ -vector space tensor product $V_1 \otimes V_2$ is a $\mathbb{C}(G_1 \times G_2)$ -module with action given by $(s_1, s_2)(v_1 \otimes v_2) = (s_1 v_1) \otimes (s_2 v_2)$ ($s_i \in G_i, v_i \in V_i$). If both groups G_1 and G_2 equal the same group G , then $V_1 \otimes V_2$ is a $\mathbb{C}G$ -module with action given by $s(v_1 \otimes v_2) = (sv_1) \otimes (sv_2)$ ($s \in G, v_i \in V_i$).
7. *Contragredient*: If V is a $\mathbb{C}G$ -module, then the $\mathbb{C}$ -vector space dual V^* is a $\mathbb{C}G$ -module (called the **contragredient** of V) with action given by $(sf)(v) = f(s^{-1}v)$ ($s \in G, f \in V^*, v \in V$).

68.2 Matrix Representations

Throughout, G denotes a finite group, e denotes its identity element, and V denotes a finite dimensional complex vector space.

Definitions:

A **matrix representation** of G of **degree** n (over the field $\mathbb{C}$) is a homomorphism $R : G \rightarrow GL_n(\mathbb{C})$, where $GL_n(\mathbb{C})$ is the group of nonsingular $n \times n$ matrices over the field $\mathbb{C}$. (For the relationship between representations and matrix representations, see the facts below.)

The **empty matrix** is a 0×0 matrix having no entries. The trace of the empty matrix is 0. $GL_0(\mathbb{C})$ is the trivial group whose only element is the empty matrix.

Two matrix representations R and R' are **equivalent** (or **isomorphic**) if they have the same degree, say n , and there exists a nonsingular $n \times n$ matrix P such that $R'(s) = PR(s)P^{-1}$ for all $s \in G$.

A matrix representation of G is **reducible** if it is equivalent to a matrix representation R having the property that for each $s \in G$, the matrix $R(s)$ has the block form

$$R(s) = \begin{bmatrix} X(s) & Z(s) \\ 0 & Y(s) \end{bmatrix}$$

(block sizes independent of s).

A matrix representation is **irreducible** if it has nonzero degree and it is not reducible.

The **kernel** of a matrix representation R of G of degree n is the set of all $s \in G$ for which $R(s) = I_n$.

A matrix representation of G is **faithful** if its kernel consists of the identity element alone.

Facts:

The following facts can be found in [Isa94, pp. 10–11, 32] or [Ser77, pp. 11–14].

1. If R is a matrix representation of G , then
 - $R(e) = I$,
 - $R(st) = R(s)R(t)$ for all $s, t \in G$,
 - $R(s^{-1}) = R(s)^{-1}$ for all $s \in G$.

2. If $\rho = \rho_V$ is a representation of G of degree n and $\mathcal{B}$ is an ordered basis for V , then $R_{\rho, \mathcal{B}}(s) = [\rho(s)]_{\mathcal{B}}$ defines a matrix representation $R_{\rho, \mathcal{B}}: G \rightarrow GL_n(\mathbb{C})$ called the **matrix representation of G afforded by the representation ρ** (or **by the $\mathbb{C}G$ -module V**) **with respect to the basis $\mathcal{B}$** . Conversely, if R is a matrix representation of G of degree n and $V = \mathbb{C}^n$, then $\rho(s)(v) = R(s)v$ ($s \in G, v \in V$) defines a representation ρ of G and $R = R_{\rho, \mathcal{B}}$, where $\mathcal{B}$ is the standard ordered basis of V .
3. If R and R' are matrix representations afforded by representations ρ and ρ' , respectively, then R and R' are equivalent if and only if ρ and ρ' are equivalent. In particular, two matrix representations that are afforded by the same representation are equivalent regardless of the chosen bases.
4. If $\rho = \rho_V$ is a representation of G and W is a G -stable subspace of V and a basis for W is extended to a basis $\mathcal{B}$ of V , then for each $s \in G$ the matrix $R_{\rho, \mathcal{B}}(s)$ is of block form

$$R_{\rho, \mathcal{B}}(s) = \begin{bmatrix} X(s) & Z(s) \\ 0 & Y(s) \end{bmatrix},$$

where X and Y are the matrix representations afforded by ρ_W (with respect to the given basis) and $\rho_{V/W}$ (with respect to the induced basis), respectively.

5. If the matrix representation R of G is afforded by a representation ρ , then R is irreducible if and only if ρ is irreducible.
6. The group G is Abelian if and only if every irreducible matrix representation of G is of degree one.
7. *Maschke's Theorem (for matrix representations)*: If R is a matrix representation of G and for each $s \in G$ the matrix $R(s)$ is of block form

$$R(s) = \begin{bmatrix} X(s) & Z(s) \\ 0 & Y(s) \end{bmatrix}$$

(block sizes independent of s), then R is equivalent to the matrix representation R' given by

$$R'(s) = \begin{bmatrix} X(s) & 0 \\ 0 & Y(s) \end{bmatrix}$$

($s \in G$).

8. *Schur relations*: Let R and R' be irreducible matrix representations of G of degrees n and n' , respectively. For $1 \leq i, j \leq n$ and $1 \leq i', j' \leq n'$ define functions $r_{ij}, r'_{i'j'}: G \rightarrow \mathbb{C}$ by $R(s) = [r_{ij}(s)]$, $R'(s) = [r'_{i'j'}(s)]$ ($s \in G$).
- If R' is not equivalent to R , then for all $1 \leq i, j \leq n$ and $1 \leq i', j' \leq n'$

$$\sum_{s \in G} r_{ij}(s^{-1})r'_{i'j'}(s) = 0.$$

- For all $1 \leq i, j, k, l \leq n$

$$\sum_{s \in G} r_{ij}(s^{-1})r_{kl}(s) = \begin{cases} |G|/n & \text{if } i = l \text{ and } j = k \\ 0 & \text{otherwise.} \end{cases}$$

Examples:

1. An example of a degree two matrix representation of the symmetric group S_3 is given by

$$\begin{aligned} R(e) &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, & R(12) &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, & R(23) &= \begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix}, \\ R(13) &= \begin{bmatrix} 1 & 0 \\ -1 & -1 \end{bmatrix}, & R(123) &= \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, & R(132) &= \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}. \end{aligned}$$

(Cf. Example 2 of section 68.4.)

2. The matrix representation R of the additive group $\mathbf{Z}_3$ of integers modulo 3 afforded by the regular representation with respect to the basis $\mathbf{Z}_3 = \{0, 1, 2\}$ (ordered as indicated) is given by

$$R(0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R(1) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad R(2) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

3. Let $\rho : G \rightarrow GL(V)$ and $\rho' : G \rightarrow GL(V')$ be two representations of G , let $\mathcal{B}$ and $\mathcal{B}'$ be bases of V and V' , respectively, and let $R = R_{\rho, \mathcal{B}}$ and $R' = R_{\rho', \mathcal{B}'}$ be the afforded matrix representations.
- The matrix representation afforded by the direct sum $V \oplus V'$ with respect to the basis $\{(b, 0), (0, b') \mid b \in \mathcal{B}, b' \in \mathcal{B}'\}$ is given by $s \mapsto R(s) \oplus R'(s)$ (direct sum of matrices).
 - The matrix representation afforded by the tensor product $V \otimes V'$ with respect to the basis $\{b \otimes b' \mid b \in \mathcal{B}, b' \in \mathcal{B}'\}$ is given by $s \mapsto R(s) \otimes R'(s)$ (Kronecker product of matrices).
 - The matrix representation afforded by the contragredient V^* with respect to the dual basis of $\mathcal{B}$ is given by $s \mapsto (R(s)^{-1})^T$ (inverse transpose of matrix).

68.3 Characters

Throughout, G denotes a finite group, e denotes its identity element, and V denotes a finite dimensional complex vector space.

Definitions:

The **character of G afforded by a matrix representation** R of G is the function $\chi : G \rightarrow \mathbb{C}$ defined by $\chi(s) = \text{tr } R(s)$.

The **character of G afforded by a representation** $\rho = \rho_V$ of G is the character afforded by the corresponding matrix representation $R_{\rho, \mathcal{B}}$, where $\mathcal{B}$ is a basis for V .

The **character of G afforded by a $\mathbb{C}G$ -module** V is the character afforded by the corresponding representation ρ_V .

An **irreducible character** is a character afforded by an irreducible representation.

The **degree** of a character χ of G is the number $\chi(e)$.

A **linear character** is a character of degree one.

If χ_1 and χ_2 are two characters of G , their **sum** is defined by $(\chi_1 + \chi_2)(s) = \chi_1(s) + \chi_2(s)$ and their **product** is defined by $(\chi_1 \chi_2)(s) = \chi_1(s) \chi_2(s)$ ($s \in G$).

If χ is a character of G , its **complex conjugate** is defined by $\overline{\chi}(s) = \overline{\chi(s)}$ ($s \in G$), where $\overline{\chi(s)}$ denotes the conjugate of the complex number $\chi(s)$.

The **kernel** of a character χ of G is the set $\{s \in G \mid \chi(s) = \chi(e)\}$.

A character χ of G is **faithful** if its kernel consists of the identity element alone.

The **principal character** of G is the character 1_G satisfying $1_G(s) = 1$ for all $s \in G$.

The **zero character** of G is the character 0_G satisfying $0_G(s) = 0$ for all $s \in G$.

If χ and ψ are two characters of G , then ψ is called a **constituent** of χ if $\chi = \psi + \psi'$ with ψ' a character (possibly zero) of G .

Facts:

The following facts can be found in [Isa94, pp. 14–23, 38–40, 59] or [Ser77, pp. 10–19, 27, 52].

1. The principal character of G is the character afforded by the representation of G that maps every $s \in G$ to $[1] \in GL_1(\mathbb{C})$. The zero character of G is the character afforded by the representation of G that maps every $s \in G$ to the empty matrix in $GL_0(\mathbb{C})$.

2. The degree of a character of G equals the dimension of V , where ρ_V is a representation affording the character.
3. If χ is a character of G , then $\chi(s^{-1}) = \overline{\chi(s)}$ and $\chi(t^{-1}st) = \chi(s)$ for all $s, t \in G$.
4. Two characters of G are equal if and only if representations affording them are equivalent.
5. The number of distinct irreducible characters of G is the same as the number of conjugacy classes of G .
6. Every character χ of G can be expressed in the form $\chi = \sum_{\varphi \in Irr(G)} m_{\varphi} \varphi$, where $Irr(G)$ denotes the set of irreducible characters of G and where each m_{φ} is a nonnegative integer (called the **multiplicity** of φ as a constituent of χ).
7. A nonzero character of G is irreducible if and only if it is not the sum of two nonzero characters of G .
8. The kernel of a character equals the kernel of a representation affording the character.
9. The degree of an irreducible character of G divides the order of G .
10. A character of G is linear if and only if it is a homomorphism from G into the multiplicative group of nonzero complex numbers under multiplication.
11. The group G is abelian if and only if every irreducible character of G is linear.
12. The sum of the squares of the irreducible character degrees equals the order of G .
13. *Irreducible characters of direct products:* Let G_1 and G_2 be finite groups. Denoting by $Irr(G)$ the set of irreducible characters of the group G , we have $Irr(G_1 \times G_2) = Irr(G_1) \times Irr(G_2)$, where an element (χ_1, χ_2) of the Cartesian product on the right is viewed as a function on the direct product $G_1 \times G_2$ via $(\chi_1, \chi_2)(s_1, s_2) = \chi_1(s_1)\chi_2(s_2)$.
14. *Burnside's Vanishing Theorem:* If χ is a nonlinear irreducible character of G , then $\chi(s) = 0$ for some $s \in G$.

Examples:

See also examples in the next section.

1. If V_1 and V_2 are $\mathbb{C}G$ -modules and χ_1 and χ_2 , respectively, are the characters of G they afford, then the direct sum $V_1 \oplus V_2$ affords the sum $\chi_1 + \chi_2$ and the tensor product $V_1 \otimes V_2$ affords the product $\chi_1 \chi_2$.
2. If V is a $\mathbb{C}G$ -module and χ is the character it affords, then the contragredient V^* affords the complex conjugate character $\overline{\chi}$.
3. Let X be a finite set on which an action of G is given and let ρ be the corresponding permutation representation of G (see Example 3 of Section 68.1). If χ is the character afforded by ρ , then for each $s \in G$, $\chi(s)$ is the number of ones on the main diagonal of the permutation matrix $[\rho(s)]_X$, which is the same as the number of fixed points of X under the action of s : $\chi(s) = |\{x \in X \mid sx = x\}|$. The matrix representation of $\mathbf{Z}_3$ given in Example 2 of Section 68.2 is afforded by a permutation representation, namely, the regular representation; it affords the character χ given by $\chi(0) = 3$, $\chi(1) = 0$, $\chi(2) = 0$ in accordance with the statement above.

68.4 Orthogonality Relations and Character Table

Throughout, G denotes a finite group, e denotes its identity element, and V denotes a finite dimensional complex vector space.

Definitions:

A function $f: G \rightarrow \mathbb{C}$ is called a **class function** if it is constant on the conjugacy classes of G , that is, if $f(t^{-1}st) = f(s)$ for all $s, t \in G$.

The **inner product** of two functions f and g from G to $\mathbb{C}$ is the complex number

$$(f, g)_G = \frac{1}{|G|} \sum_{s \in G} f(s) \overline{g(s)}.$$

The **character table** of the group G is the square array with entry in the i th row and j th column equal to the complex number $\chi_i(c_j)$, where $\text{Irr}(G) = \{\chi_1, \dots, \chi_k\}$ is the set of distinct irreducible characters of G and $\{c_1, \dots, c_k\}$ is a set consisting of exactly one element from each conjugacy class of G .

Facts:

The following facts can be found in [Isa94, pp. 14–21, 30] or [Ser77, pp.10–19].

1. Each character of G is a class function.
2. *First Orthogonality Relation:* If φ and ψ are two irreducible characters of G , then

$$(\phi, \psi)_G = \frac{1}{|G|} \sum_{s \in G} \varphi(s) \overline{\psi(s)} = \begin{cases} 1 & \text{if } \varphi = \psi \\ 0 & \text{if } \varphi \neq \psi. \end{cases}$$

3. *Second Orthogonality Relation:* If s and t are two elements of G , then

$$\sum_{\chi \in \text{Irr}(G)} \chi(s) \overline{\chi(t)} = \begin{cases} |G|/c(s) & \text{if } t \text{ is conjugate to } s \\ 0 & \text{if } t \text{ is not conjugate to } s, \end{cases}$$

where $c(s)$ denotes the number of elements in the conjugacy class of s .

4. *Generalized Orthogonality Relation:* If φ and ψ are two irreducible characters of G and t is an element of G , then

$$\frac{1}{|G|} \sum_{s \in G} \varphi(st) \overline{\psi(s)} = \begin{cases} \varphi(t)/\varphi(e) & \text{if } \varphi = \psi \\ 0 & \text{if } \varphi \neq \psi. \end{cases}$$

This generalizes the First Orthogonality Relation (Fact 2).

5. The set of complex-valued functions on G is a complex inner product space with inner product as defined above. The set of class functions on G is a subspace.
6. A character χ of G is irreducible if and only if $(\chi, \chi)_G = 1$.
7. The set $\text{Irr}(G)$ of irreducible characters of G is an orthonormal basis for the inner product space of class functions on G .
8. If the character χ of G is expressed as a sum of irreducible characters (see Fact 6 of Section 68.3), then the number of times the irreducible character φ appears as a summand is $(\chi, \varphi)_G$. In particular, $\varphi \in \text{Irr}(G)$ is a constituent of χ if and only if $(\chi, \varphi)_G \neq 0$.
9. Isomorphic groups have identical character tables (up to a reordering of rows and columns). The converse of this statement does not hold since, for example, the dihedral group and the quaternion group (both of order eight) have the same character table, yet they are not isomorphic.

Examples:

1. The character table of the group $\mathbf{Z}_4$ of integers modulo four is

	0	1	2	3
χ_0	1	1	1	1
χ_1	1	i	-1	$-i$
χ_2	1	-1	1	-1
χ_3	1	$-i$	-1	i

2. The character table of the symmetric group S_3 is

	(1)	(12)	(123)
χ_0	1	1	1
χ_1	1	-1	1
χ_2	2	0	-1

Note that χ_2 is the character afforded by the matrix representation of S_3 given in Example 1 of Section 68.2.

3. The character table of the symmetric group S_4 is

	(1)	(12)	(12)(34)	(123)	(1234)
χ_0	1	1	1	1	1
χ_1	1	-1	1	1	-1
χ_2	2	0	2	-1	0
χ_3	3	1	-1	0	-1
χ_4	3	-1	-1	0	1

4. The character table of the alternating group A_4 is

	(1)	(12)(34)	(123)	(132)
χ_0	1	1	1	1
χ_1	1	1	ω	ω^2
χ_2	1	1	ω^2	ω
χ_3	3	-1	0	0

where $\omega = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

5. Let ρ_V be a representation of G and for each irreducible character φ of G put

$$T_\varphi = \frac{\varphi(e)}{|G|} \sum_{s \in G} \varphi(s^{-1}) \rho_V(s) : V \rightarrow V.$$

Then the Generalized Orthogonality Relation (Fact 4) shows that

$$T_\varphi T_\psi = \begin{cases} T_\varphi & \text{if } \varphi = \psi \\ 0 & \text{if } \varphi \neq \psi \end{cases}$$

and

$$\sum_{\varphi \in \text{Irr}(G)} T_\varphi = 1_V,$$

where 1_V denotes the identity operator on V . Moreover, $V = \bigoplus_{\varphi \in \text{Irr}(G)} T_\varphi(V)$ (internal direct sum).

68.5 Restriction and Induction of Characters

Throughout, G denotes a finite group, e denotes its identity element, and V denotes a finite dimensional complex vector space.

Definitions:

If χ is a character of G and H is a subgroup of G , then the **restriction** of χ to H is the character χ_H of H obtained by restricting the domain of χ .

A character φ of a subgroup H of G is **extendible** to G if $\varphi = \chi_H$ for some character χ of G .

If φ is a character of a subgroup H of G , then the **induced character** from H to G is the character φ^G of G given by the formula

$$\varphi^G(s) = \frac{1}{|H|} \sum_{t \in G} \varphi^\circ(t^{-1}st),$$

where φ° is defined by $\varphi^\circ(x) = \varphi(x)$ if $x \in H$ and $\varphi^\circ(x) = 0$ if $x \notin H$.

If φ is a character of a subgroup H of G and s is an element of G , then the **conjugate character** of φ by s is the character φ^s of $H^s = s^{-1}Hs$ given by $\varphi^s(h^s) = \varphi(h)$ ($h \in H$), where $h^s = s^{-1}hs$.

Facts:

The following facts can be found in [Isa94, pp. 62–63, 73–79] or [Ser77, pp. 55–58].

1. The restricted character defined above is indeed a character: χ_H is afforded by the restriction to H of a representation affording χ .
2. The induced character defined above is indeed a character: Let V be a $\mathbb{C}H$ -module affording φ and put $V^G = \bigoplus_{t \in T} V_t$, where $G = \bigcup_{t \in T} tH$ (disjoint union) and $V_t = V$ for each t . Then V^G is a $\mathbb{C}G$ -module that affords φ^G , where the action is given as follows: For $s \in G$ and $v \in V_t$, sv is the element $h'v$ of $V_{t'}$, where $st = t'h$ ($t' \in T, h \in H$).
3. The conjugate character defined above is indeed a character: φ^s is afforded by the representation of H^s obtained by composing the homomorphism $H^s \rightarrow H$, $h^s \mapsto h$ with a representation affording φ .
4. *Additivity of restriction:* Let H be a subgroup of G . If χ and χ' are characters of G , then $(\chi + \chi')_H = \chi_H + \chi'_H$.
5. *Additivity of induction:* Let H be a subgroup of G . If φ and φ' are characters of H , then $(\varphi + \varphi')^G = \varphi^G + \varphi'^G$.
6. *Transitivity of induction:* Let H and K be subgroups of G with $H \subseteq K$. If φ is a character of H , then $(\varphi^K)^G = \varphi^G$.
7. *Degree of induced character:* If H is a subgroup of G and φ is a character of H , then the degree of the induced character φ^G equals the product of the index of H in G and the degree of φ : $\varphi^G(e) = [G : H]\varphi(e)$.
8. Let χ be a character of G and let H be a subgroup of G . If the restriction χ_H is irreducible, then so is χ . The converse of this statement does not hold. In fact, if H is the trivial subgroup then $\chi_H = \chi(e)1_H$, so any nonlinear irreducible character (e.g., χ_2 in Example 2 of section 68.4) provides a counterexample.
9. Let H be a subgroup of G and let φ be a character of H . If the induced character φ^G is irreducible, then so is φ . The converse of this statement does not hold (see Example 3).
10. Let H be a subgroup of G . If φ is an irreducible character of H , then there exists an irreducible character χ of G such that φ is a constituent of χ_H .
11. *Frobenius Reciprocity:* If χ is a character of G and φ is a character of a subgroup H of G , then $(\varphi^G, \chi)_G = (\varphi, \chi_H)_H$.
12. If χ is a character of G and φ is a character of a subgroup H of G , then $(\varphi \chi_H)^G = \varphi^G \chi$.
13. *Mackey's Subgroup Theorem:* If H and K are subgroups of G and φ is a character of H , then $(\varphi^G)_K = \sum_{t \in T} (\varphi_{H' \cap K}^t)^K$, where T is a set of representatives for the (H, K) -double cosets in G (so that $G = \bigcup_{t \in T} HtK$, a disjoint union).
14. If φ is a character of a normal subgroup N of G , then for each $s \in G$, the conjugate φ^s is a character of N . Moreover, $\varphi^s(n) = \varphi(sns^{-1})$ ($n \in N$).
15. *Clifford's Theorem:* Let N be a normal subgroup of G , let χ be an irreducible character of G , and let φ be an irreducible constituent of χ_N . Then $\chi_N = m \sum_{i=1}^h \varphi_i$, where $\varphi_1, \dots, \varphi_h$ are the distinct conjugates of φ under the action of G and $m = (\chi_N, \varphi)_N$.

Examples:

1. Given a subgroup H of G , the induced character $(1_H)^G$ equals the permutation character corresponding to the action of G on the set of left cosets of H in G given by $s(tH) = (st)H$ ($s, t \in G$).
2. The induced character $(1_{\{e\}})^G$ equals the permutation character corresponding to the action of G on itself given by left multiplication. It is the character of the (left) regular representation of G . This character satisfies

$$(1_{\{e\}})^G(s) = \begin{cases} |G| & \text{if } s = e \\ 0 & \text{if } s \neq e. \end{cases}$$

3. As an illustration of Frobenius Reciprocity (Fact 11), we have $((1_{\{e\}})^G, \chi)_G = (1_{\{e\}}, \chi_{\{e\}})_{\{e\}} = \chi(e)$ for any irreducible character χ of G . Hence, $(1_{\{e\}})^G = \sum_{\chi \in \text{Irr}(G)} \chi(e)\chi$ (cf. Fact 8 of section 68.4), that is, in the character of the regular representation (see Example 2), each irreducible character appears as a constituent with multiplicity equal to its degree.

68.6 Representations of the Symmetric Group

Definitions:

Given a natural number n , a tuple $\alpha = [\alpha_1, \dots, \alpha_h]$ of nonnegative integers is a (**proper**) **partition** of n (written $\alpha \vdash n$) provided

- $\alpha_i \geq \alpha_{i+1}$ for all $1 \leq i < h$,
- $\sum_{i=1}^h \alpha_i = n$.

The **conjugate partition** of a partition $\alpha \vdash n$ is the partition $\alpha' \vdash n$ with i th component α'_i equal to the number of indices j for which $\alpha_j \geq i$. This partition is also called the **partition associated with α** .

Given two partitions $\alpha = [\alpha_1, \dots, \alpha_h]$ and $\beta = [\beta_1, \dots, \beta_k]$ of n , α **majorizes** (or **dominates**) β if

$$\sum_{i=1}^j \alpha_i \geq \sum_{i=1}^j \beta_i$$

for each $1 \leq j \leq h$. This is expressed by writing $\alpha \succeq \beta$ (or $\beta \preceq \alpha$).

The **Young subgroup** of the symmetric group S_n corresponding to a partition $\alpha = [\alpha_1, \dots, \alpha_h]$ of n is the internal direct product $S_\alpha = S_{A_1} \times \cdots \times S_{A_h}$, where S_{A_i} is the subgroup of S_n consisting of those permutations that fix every integer *not* in the set

$$A_i = \left\{ 1 \leq k \leq n \mid \sum_{j=1}^{i-1} \alpha_j < k \leq \sum_{j=1}^i \alpha_j \right\}$$

(an empty sum being interpreted as zero).

The **alternating character** of the symmetric group S_n is the character ϵ_n given by

$$\epsilon_n(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is even,} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases}$$

Let G be a subgroup of S_n and let χ be a character of G . The **generalized matrix function** $d_\chi : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}$ is defined by

$$d_\chi(A) = \sum_{s \in G} \chi(s) \prod_{j=1}^n a_{js(j)}.$$

When $G = S_n$ and χ is irreducible, d_χ is called an **immanant**.

Facts:

The following facts can be found in [JK81, pp. 15, 35–37] or [Mer97, pp. 99–103, 214].

1. If χ is a character of the symmetric group S_n , then $\chi(\sigma)$ is an integer for each $\sigma \in S_n$.
2. *Irreducible character associated with a partition:* Given a partition α of n , there is a unique irreducible character χ_α that is a constituent of both the induced character $(1_{S_\alpha})^{S_n}$ and the induced character $((\epsilon_n)_{S_{\alpha'}})^{S_n}$. The map $\alpha \mapsto \chi_\alpha$ defines a bijection from the set of partitions of n to the set $\text{Irr}(S_n)$ of irreducible characters of S_n .
3. If α and β are partitions of n , then the irreducible character χ_α is a constituent of the induced character $(1_{S_\beta})^{S_n}$ if and only if α majorizes β .
4. If α is a partition of n , then $\chi_{\alpha'} = \epsilon_n \chi_\alpha$.
5. *Schur's inequality:* Let χ be an irreducible character of a subgroup G of S_n . For any positive semidefinite matrix $A \in \mathbb{C}^{n \times n}$, $d_\chi(A)/\chi(e) \geq \det A$.

Examples:

1. $\alpha = [5, 3^2, 2, 1^3]$ (meaning $[5, 3, 3, 2, 1, 1, 1]$) is a partition of 16. Its conjugate is $\alpha' = [7, 4, 3, 1, 1]$.
2. $\chi_{[n]} = 1_{S_n}$ and $\chi_{[1^n]} = \epsilon_n$.
3. In the notation of Example 3 of section 68.4, we have $\chi_0 = \chi_{[4]}$, $\chi_1 = \chi_{[1^4]}$, $\chi_2 = \chi_{[2^2]}$, $\chi_3 = \chi_{[3,1]}$, and $\chi_4 = \chi_{[2,1^2]}$.
4. According to Fact 4, a partition α of n is self-conjugate (meaning $\alpha' = \alpha$) if and only if $\chi_\alpha(\sigma) = 0$ for every odd permutation $\sigma \in S_n$.
5. As an illustration of Fact 3, we have

$$(1_{S_{[2,1^2]}})^{S_4} = \chi_{[4]} + \chi_{[3,1]} + \chi_{[3,1]} + \chi_{[2^2]} + \chi_{[2,1^2]}.$$

The irreducible constituents of the induced character $(1_{S_{[2,1^2]}})^{S_4}$ are the terms on the right-hand side of the equation. Note that $[4]$, $[3, 1]$, $[2^2]$, and $[2, 1^2]$ are precisely the partitions of 4 that majorize $[2, 1^2]$ in accordance with the fact.

6. When $G = S_n$ and $\chi = \epsilon_n$ (the alternating character), $d_\chi(A)$ is the determinant of $A \in \mathbb{C}^{n \times n}$.
7. When $G = S_n$ and $\chi = 1_G$ (the principal character), $d_\chi(A) = \sum_{s \in G} \prod_{j=1}^n a_{js(j)}$ is called the **permanent** of $A \in \mathbb{C}^{n \times n}$, denoted *per* A .
8. The following open problem is known as the **Permanental Dominance** (or **Permanent-on-Top**) **Conjecture**: Let χ be an irreducible character of a subgroup G of S_n . For any positive semidefinite matrix $A \in \mathbb{C}^{n \times n}$, *per* $A \geq d_\chi(A)/\chi(e)$ (cf. Fact 5 and Examples 6 and 7).

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69

Nonassociative Algebras

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One of the earliest surveys on nonassociative algebras is the article by Shirshov [Shi58] that introduced the phrase “rings that are nearly associative.” The first book in the English language devoted to a systematic study of nonassociative algebras is Schafer [Sch66]. A comprehensive exposition of the work of the Russian school is Zhevlakov, Slinko, Shestakov, and Shirshov [ZSS82]. A collection of open research problems in algebra, including many problems on nonassociative algebra, is the *Dniester Notebook* [FKS93]; the survey article by Kuzmin and Shestakov [KS95] is from the same period. Three books on Jordan algebras that contain substantial material on general nonassociative algebras are Braun and Koecher [BK66], Jacobson [Jac68], and McCrimmon [McC04]. Recent research appears in the *Proceedings of the International Conferences on Nonassociative Algebra and Its Applications* [Gon94], [CGG00], [SSS06]. The present chapter provides very limited information on Lie algebras, since they are the subject of Chapter 70. The last section (Section 69.9) presents three applications of computational linear algebra to the study of polynomial identities for nonassociative algebras: Pseudorandom vectors in a nonassociative algebra, the expansion matrix for a nonassociative operation, and the representation theory of the symmetric group.

69.1 Introduction

Definitions:

An **algebra** is a vector space A over a field F together with a bilinear **multiplication** $(x, y) \mapsto xy$ from $A \times A$ to A ; that is, **distributivity** holds for all $a, b \in F$ and all $x, y, z \in A$:

$$(ax + by)z = a(xz) + b(yz), \quad x(ay + bz) = a(xy) + b(xz).$$

The **dimension** of an algebra A is its dimension as a vector space.

An algebra A is **finite dimensional** if A is a finite dimensional vector space.

The **structure constants** of a finite dimensional algebra A over F with basis $\{x_1, \dots, x_n\}$ are the scalars $c_{ij}^k \in F$ ($i, j, k = 1, \dots, n$) defined by:

$$x_i x_j = \sum_{k=1}^n c_{ij}^k x_k.$$

An algebra A is **unital** if there exists an element $1 \in A$ for which

$$1x = x1 = x \quad \text{for all } x \in A.$$

An **involution** of the algebra A is a linear mapping $j: A \rightarrow A$ satisfying

$$j(j(x)) = x \quad \text{and} \quad j(xy) = j(y)j(x) \quad \text{for all } x, y \in A.$$

An algebra A is a **division algebra** if for every $x, y \in A$ with $x \neq 0$ the equations $xv = y$ and $wx = y$ are solvable in A .

The **associator** in an algebra is the trilinear function

$$(x, y, z) = (xy)z - x(yz).$$

An algebra A is **associative** if the associator vanishes identically:

$$(x, y, z) = 0 \quad \text{for all } x, y, z \in A.$$

An algebra is **nonassociative** if the above identity is not necessarily satisfied.

An algebra A is **alternative** if it satisfies the **right and left alternative identities**

$$(y, x, x) = 0 \quad \text{and} \quad (x, x, y) = 0 \quad \text{for all } x, y \in A.$$

An algebra A is **anticommutative** if it satisfies the identity

$$x^2 = 0 \quad \text{for all } x \in A.$$

(This implies that $xy = -yx$, and the converse holds in characteristic $\neq 2$.)

The **Jacobian** in an anticommutative algebra is defined by

$$J(x, y, z) = (xy)z + (yz)x + (zx)y.$$

A **Lie algebra** is an anticommutative algebra satisfying the **Jacobi identity**

$$J(x, y, z) = 0 \quad \text{for all } x, y, z \in A.$$

A **Malcev algebra** is an anticommutative algebra satisfying the identity

$$J(x, y, xz) = J(x, y, z)x \quad \text{for all } x, y, z \in A.$$

The **commutator** in an algebra A is the bilinear function

$$[x, y] = xy - yx.$$

The **minus algebra** A^- of an algebra A is the algebra with the same underlying vector space as A but with $[x, y]$ as the multiplication.

An algebra A is **commutative** if it satisfies the identity

$$xy = yx \quad \text{for all } x, y \in A.$$

A **Jordan algebra** is a commutative algebra satisfying the **Jordan identity**

$$(x^2, y, x) = 0 \quad \text{for all } x, y \in A.$$

The **Jordan product** (or **anticommutator**) in an algebra A is the bilinear function

$$x * y = xy + yx.$$

(The notation $x \circ y$ is also common.)

The **plus algebra** A^+ of an algebra A over a field F of characteristic $\neq 2$ is the algebra with the same underlying vector space as A but with $x \cdot y = \frac{1}{2}(x * y)$ as the multiplication.

A Jordan algebra is called **special** if it is isomorphic to a subalgebra of A^+ for some associative algebra A ; otherwise it is called **exceptional**.

Given two algebras A and B over a field F , a **homomorphism** from A to B is a linear mapping $f: A \rightarrow B$ that satisfies $f(xy) = f(x)f(y)$ for all $x, y \in A$.

An **isomorphism** is a homomorphism that is a linear isomorphism of vector spaces.

Let A be an algebra. Given two subsets $B, C \subseteq A$ we write BC for the **subspace spanned by the products** yz where $y \in B, z \in C$.

A **subalgebra** of A is a subspace B satisfying $BB \subseteq B$.

The **subalgebra generated by a set** $S \subseteq A$ is the smallest subalgebra of A containing S .

A (two-sided) **ideal** of an algebra A is a subalgebra B satisfying $AB + BA \subseteq B$.

Given two algebras A and B over the field F , the **(external) direct sum** of A and B is the vector space direct sum $A \oplus B$ with the multiplication

$$(w, x)(y, z) = (wy, xz) \quad \text{for all } w, y \in A \quad \text{and all } x, z \in B.$$

Given an algebra A with two ideals B and C , we say that A is the **(internal) direct sum** of B and C if $A = B \oplus C$ (direct sum of subspaces).

Facts: ([Shi58], [Sch66], [ZSS82], [KS95])

1. Every finite dimensional associative algebra over a field F is isomorphic to a subalgebra of a matrix algebra $F^{n \times n}$ for some n .
2. The algebra A^- is always anticommutative. If A is associative, then A^- is a Lie algebra.
3. (Poincaré–Birkhoff–Witt Theorem or PBW Theorem) Every Lie algebra is isomorphic to a subalgebra of A^- for some associative algebra A .
4. The algebra A^+ is always commutative. If A is associative, then A^+ is a Jordan algebra. (See Example 2 in Section 69.9.) If A is alternative, then A^+ is a Jordan algebra.
5. The analogue of the PBW theorem for Jordan algebras is false: Not every Jordan algebra is special. (See Example 4 below.)
6. Every associative algebra is alternative.
7. (Artin's Theorem) An algebra is alternative if and only if every subalgebra generated by two elements is associative.
8. Every Lie algebra is a Malcev algebra.
9. Every Malcev algebra generated by two elements is a Lie algebra.
10. If A is an alternative algebra, then A^- is a Malcev algebra. (See Example 3 in Section 69.9.)
11. In an external direct sum of algebras, the summands are ideals.

Examples:

1. Associativity is satisfied when the elements of the algebra are mappings of a set into itself with the composition of mappings taken as multiplication. Such is the multiplication in the algebra $\text{End } V$, the algebra of linear operators on the vector space V . Every associative algebra is isomorphic to a subalgebra of the algebra $\text{End } V$, for some V . Thus, the condition of associativity of multiplication

characterizes the **algebras of linear operators**. (Note that $\text{End } V$ is also denoted $L(V, V)$ elsewhere in this book, but $\text{End } V$ is the standard notation in the study of algebras.)

2. **Cayley–Dickson doubling process.** Let A be a unital algebra over F with an involution $x \mapsto \bar{x}$ satisfying

$$x + \bar{x}, x\bar{x} \in F \quad \text{for all } x \in A. \quad (69.1)$$

Let $a \in F, a \neq 0$. The algebra (A, a) is defined as follows: The underlying vector space is $A \oplus A$, addition and scalar multiplication are defined by the vector space formulas

$$(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2), \quad c(x_1, x_2) = (cx_1, cx_2) \quad \text{for all } c \in F, \quad (69.2)$$

and multiplication is defined by the formula

$$(x_1, x_2)(y_1, y_2) = (x_1y_1 + ay_2\bar{x}_2, \bar{x}_1y_2 + y_1x_2). \quad (69.3)$$

This algebra has an involution defined by

$$\overline{(x_1, x_2)} = (\bar{x}_1, -x_2). \quad (69.4)$$

In particular, starting with a field F of characteristic $\neq 2$, we obtain the following examples:

- (a) The algebra $\mathbb{C}(a) = (F, a)$ is commutative and associative. If the polynomial $x^2 + a$ is irreducible over F , then $\mathbb{C}(a)$ is a field, otherwise $\mathbb{C}(a) \cong F \oplus F$ (algebra direct sum).
- (b) The algebra $\mathbb{H}(a, b) = (\mathbb{C}(a), b)$ is an algebra of **generalized quaternions**, which is associative but not commutative.
- (c) The algebra $\mathbb{O}(a, b, c) = (\mathbb{H}(a, b), c)$ is an algebra of **generalized octonions** or a **Cayley–Dickson algebra**, which is alternative but not associative. (See Example 1 in Section 69.9.)

The algebras of generalized quaternions and octonions may also be defined over a field of characteristic 2 (see [Sch66], [ZSS82]).

- 3. **Real division algebras** [EHH91, Part B]. In the previous example, taking F to be the field $\mathbb{R}$ of real numbers and $a = b = c = -1$, we obtain the field $\mathbb{C}$ of **complex numbers**, the associative division algebra $\mathbb{H}$ of **quaternions**, and the alternative division algebra $\mathbb{O}$ of **octonions** (also known as the **Cayley numbers**). Real division algebras exist only in dimensions 1, 2, 4, and 8, but there are many other examples: The algebras $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$ with the multiplication $x \cdot y = \bar{x}\bar{y}$ are still division algebras, but they are not alternative and they are not unital.
- 4. **The Albert algebra.** Let $\mathbb{O}$ be the octonions and let $M_3(\mathbb{O})$ be the algebra of 3×3 matrices over $\mathbb{O}$ with involution induced by the involution of $\mathbb{O}$, that is $(a_{ij}) \mapsto (\bar{a}_{ji})$. The subalgebra $H_3(\mathbb{O})$ of Hermitian matrices in $M_3(\mathbb{O})^+$ is an exceptional Jordan algebra, the Albert algebra: There is no associative algebra A such that $H_3(\mathbb{O})$ is isomorphic to a subalgebra of A^+ .

69.2 General Properties

Definitions:

Given an algebra A and an ideal I , the **quotient algebra** A/I is the quotient space A/I with multiplication defined by $(x + I)(y + I) = xy + I$ for all $x, y \in A$.

The algebra A is **simple** if $AA \neq \{0\}$ and A has no ideals apart from $\{0\}$ and A .

The algebra A is **semisimple** if it is the direct sum of simple algebras. (The definition of semisimple that is used in the theory of Lie algebras is different; see Chapter 70.)

Set $A^1 = A^{(1)} = A$, and then by induction define

$$A^{n+1} = \sum_{i+j=n+1} A^i A^j \quad \text{and} \quad A^{(n+1)} = A^{(n)} A^{(n)} \quad \text{for } n \geq 1.$$

The algebra A is **nilpotent** if $A^n = \{0\}$ for some n and **solvable** if $A^{(s)} = \{0\}$ for some s . The smallest natural number n (respectively s) with this property is the **nilpotency index** (respectively, **solvability index**) of A .

An element $x \in A$ is **nilpotent** if the subalgebra it generates is nilpotent.

A **nil algebra** (respectively **nil ideal**) is an algebra (respectively ideal) in which every element is nilpotent.

An algebra is **power associative** if every element generates an associative subalgebra.

An **idempotent** is an element $e \neq 0$ of an algebra A satisfying $e^2 = e$. Two idempotents e, f are **orthogonal** if $ef = fe = 0$.

For an algebra A over a field F , the **degree** of A is defined to be the maximal number of mutually orthogonal idempotents in the scalar extension $\overline{F} \otimes_F A$, where $\overline{F}$ is the algebraic closure of F .

The **associator ideal** $D(A)$ of the algebra A is the ideal generated by all the associators. The **associative center** or **nucleus** $N(A)$ of A is defined by

$$N(A) = \{x \in A \mid (x, A, A) = (A, x, A) = (A, A, x) = \{0\}\}.$$

The **center** $Z(A)$ of the algebra A is defined by

$$Z(A) = \{x \in N(A) \mid [x, A] = \{0\}\}.$$

The **right and left multiplication operators** by an element $x \in A$ are defined by

$$R_x: y \mapsto yx, \quad L_x: y \mapsto xy.$$

The **multiplication algebra** of the algebra A is the subalgebra $M(A)$ of the associative algebra $\text{End } A$ (of endomorphisms of the vector space A) generated by all R_x and L_x for $x \in A$.

The **right multiplication algebra** of the algebra A is the subalgebra $R(A)$ of $\text{End } A$ generated by all R_x for $x \in A$.

The **centroid** $C(A)$ of the algebra A is the centralizer of the multiplication algebra $M(A)$ in the algebra $\text{End } A$; that is,

$$C(A) = \{T \in \text{End } A \mid TR_x = R_x T = R_{Tx}, \quad TL_x = L_x T = L_{Tx}, \quad \text{for any } x \in A\}.$$

An algebra A over a field F is **central** if $C(A) = F$.

The **unital hull** $A^\sharp$ of an algebra A over a field F is defined as follows: If A is unital, then $A^\sharp = A$; and when A has no unit, we set $A^\sharp = A \oplus F$ (vector space direct sum) and define multiplication by assuming that A is a subalgebra and the unit of F is the unit of $A^\sharp$.

Let $\mathcal{M}$ be a class of algebras closed under homomorphic images. A subclass $\mathcal{R}$ of $\mathcal{M}$ is said to be **radical** if

1. $\mathcal{R}$ is closed under homomorphic images.
2. For each $A \in \mathcal{M}$ there is an ideal $\mathcal{R}(A)$ of A such that $\mathcal{R}(A) \in \mathcal{R}$ and $\mathcal{R}(A)$ contains every ideal of A contained in $\mathcal{R}$.
3. $\mathcal{R}(A/\mathcal{R}(A)) = \{0\}$.

In this case, we call the ideal $\mathcal{R}(A)$ the **$\mathcal{R}$ -radical** of A . The algebra A is said to be **$\mathcal{R}$ -semisimple** if $\mathcal{R}(A) = \{0\}$.

If the subclass Nil of nil-algebras is radical in the class $\mathcal{M}$, then the corresponding ideal $\text{Nil } A$, for $A \in \mathcal{M}$, is called the **nil radical** of A . In this case, the algebra A is called **nil-semisimple** if $\text{Nil } A = \{0\}$. By definition, $\text{Nil } A$ contains all two-sided nil-ideals of A , and the quotient algebra $A/\text{Nil } A$ is nil-semisimple, that is, $\text{Nil } (A/\text{Nil } A) = \{0\}$.

If the subclass *Nilp* of nilpotent algebras (or the subclass *Solv* of solvable algebras) is radical in the class $\mathcal{M}$, then the corresponding ideal $\text{Nilp } A$ (respectively $\text{Solv } A$), for $A \in \mathcal{M}$, is called the **nilpotent radical** (respectively the **solvable radical**) of A .

For an algebra A over F , an A -**bimodule** is a vector space M over F with bilinear mappings

$$A \times M \rightarrow M, (x, m) \mapsto xm \quad \text{and} \quad M \times A \rightarrow M, (m, x) \mapsto mx.$$

The **split null extension** $E(A, M)$ of A by M is the algebra over F with underlying vector space $A \oplus M$ and multiplication

$$(x + m)(y + n) = xy + (xn + my) \quad \text{for all } x, y \in A, m, n \in M.$$

For an algebra A , the **regular bimodule** $\text{Reg}(A)$ is the underlying vector space of A considered as an A -bimodule, interpreting mx and xm as multiplication in A .

If M is an A -bimodule, then the mappings

$$\rho(x): m \mapsto mx, \quad \lambda(x): m \mapsto xm,$$

are linear operators on M , and the mappings

$$x \mapsto \rho(x), \quad x \mapsto \lambda(x),$$

are linear mappings from A to the algebra $\text{End}_F M$. The pair (λ, ρ) is called the **birepresentation** of A associated with the bimodule M .

The notions of **sub-bimodule**, **homomorphism of bimodules**, **irreducible bimodule**, and **faithful birepresentation** are defined in the natural way. The sub-bimodules of a regular A -bimodule are exactly the two-sided ideals of A .

Facts: ([Sch66], [Jac68], [ZSS82])

1. If A is a simple algebra, then $AA = A$.
2. (Isomorphism theorems)
 - (a) If $f: A \rightarrow B$ is a homomorphism of algebras over the field F , then $A/\ker(f) \cong \text{im}(f) \subseteq B$.
 - (b) If B_1 and B_2 are ideals of the algebra A with $B_2 \subseteq B_1$, then $(A/B_2)/(B_1/B_2) \cong A/B_1$.
 - (c) If S is a subalgebra of A and B is an ideal of A , then $B \cap S$ is an ideal of S and $(B + S)/B \cong S/(B \cap S)$.
3. The algebra A is nilpotent of index n if and only if any product of n elements (with any arrangement of parentheses) equals zero, and if there exists a nonzero product of $n - 1$ elements.
4. Every nilpotent algebra is solvable; the converse is not generally true. (See Example 1 below.)
5. In any algebra A , the sum of two solvable ideals is again a solvable ideal. If A is finite-dimensional, then A contains a unique maximal solvable ideal $\text{Solv } A$, and the quotient algebra $A/\text{Solv } A$ does not contain nonzero solvable ideals. In other words, the subclass *Solv* of solvable algebras is radical in the class of all finite dimensional algebras.
6. An algebra A is associative if and only if $D(A) = \{0\}$, if and only if $N(A) = A$.
7. Every solvable associative algebra is nilpotent.
8. The subclass *Nilp* of nilpotent algebras is radical in the class of all finite dimensional associative algebras.
9. A finite dimensional associative algebra A is semisimple if and only if $\text{Nilp } A = \{0\}$.
10. The previous two facts imply that every finite dimensional associative algebra A contains a unique maximal nilpotent ideal N such that the quotient algebra A/N is isomorphic to a direct sum of simple algebras.

11. Over an algebraically closed field F , every finite dimensional simple associative algebra is isomorphic to the algebra $F^{n \times n}$ of $n \times n$ matrices over F , for some $n \geq 1$.
12. The subclass $Nilp$ is not radical in the class of finite dimensional Lie algebras. (See Example 1 below.)
13. Over a field of characteristic zero, an algebra is power associative if and only if

$$x^2x = xx^2 \quad \text{and} \quad (x^2x)x = x^2x^2 \quad \text{for all } x.$$

14. Every power associative algebra A contains a unique maximal nil ideal $Nil A$, and the quotient algebra $A/Nil A$ is nil-semisimple, that is, it does not contain nonzero nil ideals. In other words, the subclass Nil of nil-algebras is radical in the class of all power associative algebras.
15. For a finite dimensional alternative or Jordan algebra A we have $Nil A = Solv A$.
16. For finite dimensional commutative power associative algebras, the question of the equality of the nil and solvable radicals is still open, and is known as *Albert's problem*. An equivalent question is: Are there any simple finite dimensional commutative power associative nil algebras?
17. Every nil-semisimple finite dimensional commutative power associative algebra over a field of characteristic $\neq 2, 3, 5$ has a unit element and decomposes into a direct sum of simple algebras. Every such simple algebra is either a Jordan algebra or a certain algebra of degree 2 over a field of positive characteristic.
18. Direct expansion shows that these two identities are valid in every algebra:

$$\begin{aligned} x(y, z, w) + (x, y, z)w &= (xy, z, w) - (x, yz, w) + (x, y, zw), \\ [xy, z] - x[y, z] - [x, z]y &= (x, y, z) - (x, z, y) + (z, x, y). \end{aligned}$$

From these it follows that the associative center and the center are subalgebras, and

$$D(A) = (A, A, A) + (A, A, A)A = (A, A, A) + A(A, A, A).$$

19. If $z \in Z(A)$, then for any $x \in A$ we have

$$R_z R_x = R_x R_z = R_{zx} = R_{xz}.$$

20. If A is unital, then its centroid $C(A)$ is isomorphic to its center $Z(A)$. If A is simple, then $C(A)$ is a field which contains the base field F .
21. Let A be a finite dimensional algebra with multiplication algebra $M(A)$. Then
 - (a) A is nilpotent if and only if $M(A)$ is nilpotent.
 - (b) If A is semisimple, then so is $M(A)$.
 - (c) If A is simple, then so is $M(A)$, and $M(A) \cong \text{End}_{C(A)} A$.
22. An algebra A is simple if and only if the bimodule $\text{Reg}(A)$ is irreducible.
23. If A is an alternative algebra (respectively a Jordan algebra), then its unital hull $A^\#$ is also alternative (respectively Jordan).

Examples:

1. Let A be algebra with basis x, y , and multiplication given by $x^2 = y^2 = 0$ and $xy = -yx = y$. Then A is a Lie algebra and $A^{(2)} = \{0\}$ but $A^n \neq \{0\}$ for any $n \geq 1$. Thus, A is solvable but not nilpotent.
2. Let A be an algebra over a field F with basis x_1, x_2, y, z and the following nonzero products of basis elements:

$$yx_1 = ax_1y = x_2, \quad zx_2 = ax_2z = x_1,$$